

NTA

NEET 2025



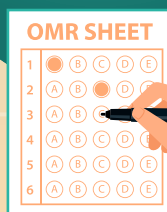
RANKERS TEST SERIES

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Test Booklet No.

ENGLISH

01

Test Booklet Code

NEET RANKERS TEST SERIES-2025



Do not open this Test Booklet until you are asked to do so.

SYLLABUS:

Physics:	Units and Measurements, Motion in a Straight Line, Motion in a Plane, Laws of Motion, Work, Energy and Power, System of Particles and Rotational Motion, Gravitation
Chemistry:	Some Basic Concepts of Chemistry, Atomic Structure, Classification of Elements & Periodicity in Properties, Chemical Bonding and Molecular Structure, Thermodynamics, Equilibrium
Botany:	The Living World, Biological classification, Plant kingdom, Morphology of flowering plants, Anatomy of flowering plants
Zoology:	Animal Kingdom, Structural organization in Animals (including Tissues, Cockroach, and Frog), Biomolecules

Important Instructions:

- The Answer Sheet is inside this Test Booklet. When you are directed to open the Test Booklet, take out the Answer Sheet and fill in the particulars on ORIGINAL Copy carefully with **blue/black** ball point pen only.
- The test is of **3 hours** duration and the Test Booklet contains **180** multiple-choice questions (four options with a single correct answer) from **Physics, Chemistry and Biology (Botany and Zoology)**.
- Each question carries 4 marks. For each correct response, the candidate will get **4 marks**. For each incorrect response, **one** mark will be deducted from the total scores. **The maximum marks are 720.**
- Use Blue/Black Ball Point Pen only** for writing particulars on this page/markings responses on Answer Sheet.
- Rough work is to be done in the space provided for this purpose in the Test Booklet only.
- On completion of the test, the candidate **must hand over the Answer Sheet (ORIGINAL and OFFICE Copy) to the Invigilator** before leaving the Room/Hall. The candidates are allowed to take away this Test Booklet with them.
- The CODE for this Booklet is 01. Make sure that the CODE printed on the Original Copy of the Answer Sheet is the same as that on this Test Booklet.** In case of discrepancy, the candidate should immediately report the matter to the Invigilator for replacement of both the Test Booklet and the Answer Sheet.
- The candidates should ensure that the Answer Sheet is not folded. Do not make any stray marks on the Answer Sheet. Do not write your Roll No. anywhere else except in the specified space in the Test Booklet/Answer Sheet.
- Use of white fluid for correction is **NOT** permissible on the Answer Sheet.
- Each candidate must show on-demand his/her Admit Card to the Invigilator
- No candidate, without special permission of the centre Superintendent or Invigilator, would leave his/her seat.
- The candidates should not leave the Examination Hall without handing over their Answer Sheet to the Invigilator on duty and sign (with time) the Attendance Sheet **twice. Cases, where a candidate has not signed the Attendance Sheet second time, will be deemed not to have handed over the Answer Sheet and dealt with as an Unfair Means case.**
- Use of Electronic/Manual Calculator is prohibited.
- The candidates are governed by all Rules and Regulations of the examination with regard to their conduct in the Examination Room/Hall. All cases of unfair means will be dealt with as per the Rules and Regulations of this examination.
- No part of the Test Booklet and Answer Sheet shall be detached under any circumstances.**
- The candidates will write the Correct Test Booklet Code as given in the Test Booklet/Answer Sheet in the Attendance Sheet.
- Compensatory time of one hour five minutes will be provided for the examination of three hours duration, whether such candidate (having a physical limitation to write) uses the facility of Scribe or not.

Name of the Candidate (in Capitals): _____

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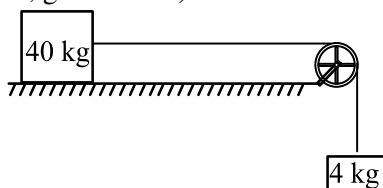
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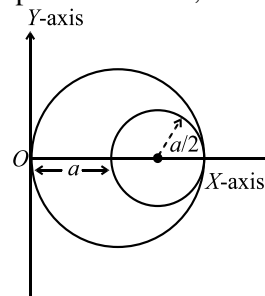
Physics

1. A block of mass 40 kg slides over a surface, when a mass of 4 kg is suspended through an inextensible massless string passing over frictionless pulley as shown below. the coefficient of kinetic friction between the surface and block is 0.02. The acceleration of block is; (Given, $g = 10 \text{ ms}^{-2}$).



- (1) 1 ms^{-2} (2) $1/5 \text{ ms}^{-2}$
(3) $4/5 \text{ ms}^{-2}$ (4) $8/11 \text{ ms}^{-2}$
2. An object of mass 1 kg is taken to a height from the surface of earth which is equal to three times the radius of earth. The gain in potential energy of the object will be;
(If, $g = 10 \text{ ms}^{-2}$ and radius of earth = 6400 km)
(1) 48 MJ (2) 24 MJ
(3) 36 MJ (4) 12 MJ
3. If vectors $\vec{A} = \cos \omega t \hat{i} + \sin \omega t \hat{j}$ and $\vec{B} = \cos\left(\frac{\omega t}{2}\right) \hat{i} + \sin\left(\frac{\omega t}{2}\right) \hat{j}$ are functions of time, then the value of t at which they are orthogonal to each other are:
(1) $t = 0$ (2) $t = \frac{\pi}{4\omega}$
(3) $t = \frac{\pi}{2\omega}$ (4) $t = \frac{\pi}{\omega}$
4. Consider a binary star system of star A and star B with masses m_A and m_B revolving in a circular orbit of radii r_A and r_B , respectively. If T_A and T_B are the time period of star A and star B, respectively, then:
(1) $\frac{T_A}{T_B} = \left(\frac{r_A}{r_B}\right)^{3/2}$ (2) $T_A > T_B$ (if $r_A > r_B$)
(3) $T_A = T_B$ (4) $T_A > T_B$ (if $m_A > m_B$)

5. A circular hole of radius $\left(\frac{a}{2}\right)$ is cut out of a circular disc of radius a as shown in figure. The centroid of the remaining circular portion with respect to point O will be;



- (1) $\frac{1}{6}a$ (2) $\frac{10}{11}a$
(3) $\frac{5}{6}a$ (4) $\frac{2}{3}a$
6. Two boys are standing at the ends A and B of a ground, where $AB = a$. The boy at B starts running in a direction perpendicular to AB with velocity v_1 . The boy at A starts running simultaneously with velocity v and catches the other boy in a time t , where t is:
(1) $\frac{a}{\sqrt{(v^2 + v_1^2)}}$ (2) $\frac{a}{\sqrt{(v^2 - v_1^2)}}$
(3) $a/(v - v_1)$ (4) $a/(v + v_1)$
7. The time dependence of the position of a particle of mass $m = 2 \text{ kg}$ is given by $r(t) = 2t\hat{i} - 3t^2\hat{j}$. Its angular momentum, with respect to the origin, at time $t = 2 \text{ sec}$ is;
(1) $36k$ (2) $-34(k - \hat{i})$
(3) $-48k$ (4) $48(\hat{i} + j)$
8. Two objects of mass 10 kg and 20 kg respectively are connected to the two ends of rigid rod of length 10 m with negligible mass. The distance of the center of mass of the system from the 10 kg mass is:
(1) 5 m (2) $\frac{10}{3} \text{ m}$
(3) $\frac{20}{3} \text{ m}$ (4) 10 m

9. A projectile is projected at 30° from horizontal with initial velocity 40 ms^{-1} . The velocity of the projectile at $t = 2 \text{ s}$ from the start will be; (Given, $g = 10 \text{ m/s}^2$)

- (1) $40\sqrt{3} \text{ ms}^{-1}$ (2) 20 ms^{-1}
(3) $20\sqrt{3} \text{ ms}^{-1}$ (4) zero

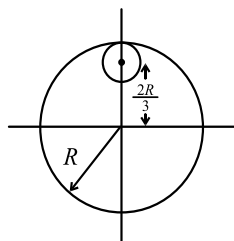
10. Match List-I with List-II.

List-I	List-II
(A) Moment of inertia of the rod (I) (length L , mass M , about an axis perpendicular to the rod passing through the mid-point)	$\frac{8ML^2}{3}$
(B) Moment of inertia of the rod (II) (length L , mass $2M$, about an axis perpendicular to the rod passing through one of its end)	$\frac{ML^2}{3}$
(C) Moment of inertia of the rod (III) (length $2L$, mass M , about an axis perpendicular to the rod passing through its midpoint)	$\frac{ML^2}{12}$
(D) Moment of inertia of the rod (IV) (length $2L$, mass $2M$, about an axis perpendicular to the rod passing through one of its end)	$\frac{2ML^2}{3}$

Choose the **correct** answer from the options given below.

- (1) A-II, B-III, C-I, D-IV
(2) A-II, B-I, C-III, D-IV
(3) A-III, B-IV, C-II, D-I
(4) A-III, B-IV, C-I, D-II

11. From a uniform circular disc of radius R and mass $9M$, a small disc of radius $R/3$ is removed as shown in the figure. The moment of inertia of the remaining disc about an axis perpendicular to the plane of the disc and passing through centre of disc is;



- (1) $4MR^2$ (2) $\frac{40}{9}MR^2$
(3) $10MR^2$ (4) $\frac{37}{9}MR^2$

12. Given below are two statements:

Statement-I: Two particles moving in the same direction do not lose all their energy in a completely inelastic collision.

Statement-II: Principle of conservation of momentum holds true for all kinds of collisions.

In the light of the above statements, choose the **most appropriate** answer from the options given below:

- (1) Statement-I is correct but Statement-II is incorrect.
(2) Statement-I is incorrect but Statement-II is correct.
(3) Both Statement-I and Statement-II are correct.
(4) Both Statement-I and Statement-II are incorrect.

13. A bomb of mass 16 kg at rest explodes into two pieces of masses 4 kg and 12 kg . The velocity of the 12 kg mass is 4 ms^{-1} . The kinetic energy of the other mass is;

- (1) 144 J (2) 288 J
(3) 192 J (4) 96 J

14. A mass m moves with a velocity v and collides inelastically with another identical mass. After collision, the 1^{st} mass moves with velocity $\frac{v}{\sqrt{3}}$ in a direction perpendicular to the initial direction of motion. Find the speed of the second mass after collision.

- (1) v (2) $\sqrt{3}v$
(3) $\frac{2}{\sqrt{3}}v$ (4) $\frac{v}{\sqrt{3}}$

15. A body at rest is moved along a horizontal straight line by a machine delivering a constant power. The distance moved by the body in time t is proportional to:

- (1) $t^{\frac{3}{2}}$ (2) $t^{\frac{1}{2}}$
(3) $t^{\frac{1}{4}}$ (4) $t^{\frac{3}{4}}$

16. If the kinetic energy of a moving body becomes four times of its initial kinetic energy, then the percentage change in its momentum will be;

- (1) 100% (2) 200%
(3) 300% (4) 400%

17. A porter lifts a heavy suitcase of mass 80 kg and at the destination lowers it down by a distance of 80 cm with a constant velocity. Calculate the work done by the porter in lowering the suitcase. [Take, $g = 9.8 \text{ ms}^{-2}$]

(1) -62720.0 J (2) -627.2 J
(3) $+627.2 \text{ J}$ (4) 784.0 J

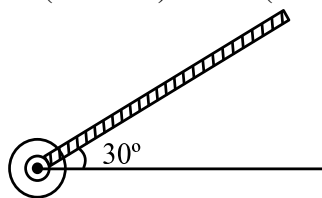
18. At any instant the velocity of a particle of mass 500 g is $(2t\hat{i} + 3t^2\hat{j})\text{ms}^{-1}$. If the force acting on the particle at $t = 1 \text{ s}$ is $(\hat{i} + x\hat{j})\text{N}$. Then the value of x will be;

(1) 3 (2) 4
(3) 6 (4) 2

19. A block of mass M placed inside a box descends vertically with acceleration a . The block exerts a force equal to one-fourth of its weight on the floor of the box. The value of a will be:

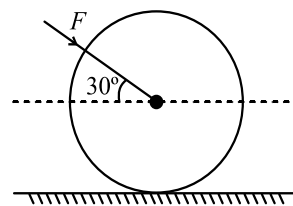
(1) $\frac{g}{4}$ (2) $\frac{g}{2}$
(3) $\frac{3g}{4}$ (4) g

20. A rod of length 50 cm is pivoted at one end. It is raised such that it makes an angle of 30° from the horizontal as shown and released from rest. Its angular speed when it passes through the horizontal (in rad s^{-1}) will be (Take, $g = 10 \text{ ms}^{-2}$)



(1) $\frac{\sqrt{30}}{2}$ (2) $\sqrt{30}$
(3) $\frac{\sqrt{20}}{3}$ (4) $\sqrt{\frac{30}{2}}$

21. As shown in figure, a 70 kg garden roller is pushed with a force of $F = 200 \text{ N}$ at an angle of 30° with horizontal. The normal reaction on the roller is (Given, $g = 10 \text{ ms}^{-2}$)



(1) $800\sqrt{2} \text{ N}$ (2) 800 N
(3) $200\sqrt{3} \text{ N}$ (4) 600 N

22. A block of mass m slides down the plane inclined at angle 30° with an acceleration $g/4$. The value of coefficient of kinetic friction will be;

(1) $\frac{2\sqrt{3}+1}{2}$ (2) $\frac{\sqrt{3}}{2}$
(3) $\frac{2\sqrt{3}-1}{2}$ (4) $\frac{1}{2\sqrt{3}}$

23. If V is the gravitational potential due to sphere of uniform density on its surface, then its value at the centre of sphere will be;

(1) $\frac{V}{2}$ (2) V
(3) $\frac{3V}{2}$ (4) $\frac{4}{3}V$

24. Let F be the force acting on a particle having position vector r and τ be the torque of this force about the origin (F , r , τ are taken as vector quantity). Then,

(1) $r \cdot \tau = 0$ and $F \cdot \tau \neq 0$
(2) $r \cdot \tau \neq 0$ and $F \cdot \tau = 0$
(3) $r \cdot \tau \neq 0$ and $F \cdot \tau \neq 0$
(4) $r \cdot \tau = 0$ and $F \cdot \tau = 0$

25. A ball is projected from the ground with a speed 15 ms^{-1} at an angle θ with horizontal so that its range and maximum height are equal, then $\tan \theta$ will be equal to-

(1) $1/4$ (2) $1/2$
(3) 2 (4) 4

26. A shell is fired from a fixed artillery gun with an initial speed u such that it hits the target on the ground at a distance R from it. If t_1 and t_2 are the values of the time taken by it to hit the target in two possible ways, the product $t_1 t_2$ is;

(1) $\frac{R}{4g}$ (2) $\frac{R}{g}$
 (3) $\frac{R}{2g}$ (4) $\frac{2R}{g}$

27. A particle is moving with velocity $\mathbf{v} = k(y\mathbf{i} + x\mathbf{j})$, where k is a constant. The general equation for its path is;

(1) $y = x^2 + \text{constant}$ (2) $y^2 = x + \text{constant}$
 (3) $xy = \text{constant}$ (4) $y^2 = x^2 + \text{constant}$

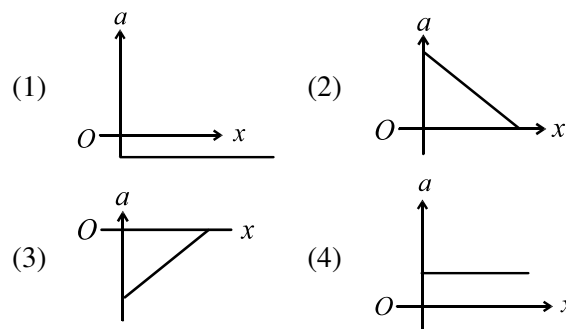
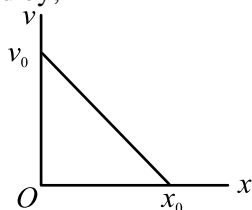
28. A particle is moving with constant speed in a circular path. When the particle turns by an angle 90° , the ratio of instantaneous velocity to its average velocity is $\pi : x\sqrt{2}$. The value of x will be;

(1) 1 (2) 2
 (3) 7 (4) 5

29. A particle is moving Eastwards with a velocity of 5 ms^{-1} . In 10 s, the velocity changes to 5 ms^{-1} Northwards. The average acceleration in this time is;

(1) $\frac{1}{\sqrt{2}} \text{ ms}^{-2}$ towards North-East
 (2) $\frac{1}{2} \text{ ms}^{-2}$ towards North
 (3) zero
 (4) $\frac{1}{\sqrt{2}} \text{ ms}^{-2}$ towards North-West

30. The velocity - displacement graph of a particle is shown in the figure. The acceleration-displacement graph of the same particle is represented by;



31. An engine of a train moving with uniform acceleration, passes the signal-post with velocity u and the last compartment with velocity v . The velocity with which middle point of the train passes the signal post is;

(1) $\sqrt{\frac{v^2 + u^2}{2}}$ (2) $\frac{v+u}{2}$
 (3) $\frac{u+v}{2}$ (4) $\sqrt{\frac{v^2 - u^2}{2}}$

32. A helicopter rises from rest on the ground vertically upwards with a constant acceleration g . A food packet is dropped from the helicopter when it is at a height h . The time taken by the packet to reach the ground is close to (Here, g is the acceleration due to gravity).

(1) $t = \frac{2}{3} \sqrt{\frac{h}{g}}$ (2) $t = 1.8 \sqrt{\frac{h}{g}}$
 (3) $t = \sqrt{\frac{2h}{3g}}$ (4) $t = 3.4 \sqrt{\frac{h}{g}}$

33. A ball is thrown vertically upward with an initial velocity of 150 m/s . The ratio of velocity after 3s and 5s is $\frac{x+1}{x}$. The value of x is;

{Take, $g = 10 \text{ m/s}^2$ }
 (1) 10 (2) 5
 (3) 6 (4) -5

34. A ball is released from a height h . If t_1 and t_2 be the time required to complete first half and second half of the distance, respectively. Then, choose the **correct** relation between t_1 and t_2 .

(1) $t_1 = (\sqrt{2})t_2$ (2) $t_1 = (\sqrt{2} - 1)t_2$
 (3) $t_2 = (\sqrt{2} + 1)t_1$ (4) $t_2 = (\sqrt{2} - 1)t_1$

35. A student measures the time period of 100 oscillations of a simple pendulum four times. The data set is 90 s, 91 s, 92 s and 95 s. If the minimum division in the measuring clock is 1 s, then the reported mean time should be;

(1) (92 ± 2) s (2) (92 ± 5) s
(3) (92 ± 1.8) s (4) (92 ± 3) s

36. The dimension of $\frac{1}{2}\epsilon_0 E^2$, where ϵ_0 is permittivity of free space and E is electric field, is:

(1) $[MLT^{-1}]$ (2) $[ML^2T^{-2}]$
(3) $[ML^{-1}T^{-2}]$ (4) $[ML^2T^{-1}]$

37. In an experiment, the percentage of error occurred in the measurement of physical quantities A , B , C and D are 1%, 2%, 3% and 4% respectively. Then the maximum percentage of error in the measurement of X , where

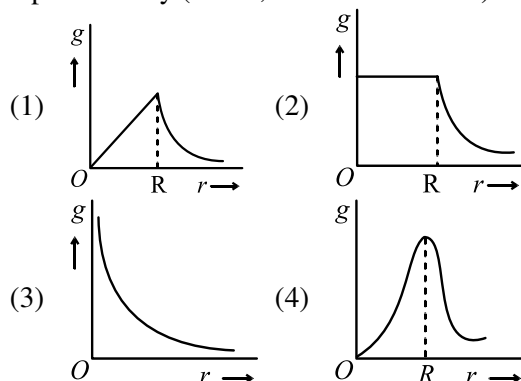
$$X = \frac{A^2 B^{1/2}}{C^{1/3} D^3} \text{ will be;}$$

(1) $\left(\frac{3}{13}\right)\%$ (2) 16 %
(3) -10 % (4) 10 %

38. If a ball is thrown vertically upwards with speed u , the distance covered during the last ' t ' seconds of its ascent is:

(1) ut (2) $\frac{1}{2}gt^2$
(3) $ut - \frac{1}{2}gt^2$ (4) $(u + gt)t$

39. The variation of acceleration due to gravity (g) with distance (r) from the centre of the earth is **correctly** represented by (Given, R = radius of earth).



40. Two blocks of masses 10 kg and 30 kg are placed on the same straight line with coordinates (0, 0) cm and $(x, 0)$ cm, respectively. The block of 10 kg is moved on the same line through a distance of 6 cm towards the other block. The distance through which the block of 30 kg must be moved to keep the position of centre of mass of the system unchanged is;

(1) 4 cm towards the 10 kg block
(2) 2 cm away from the 10 kg block
(3) 2 cm towards the 10 kg block
(4) 4 cm away from the 10 kg block

41. Given below are two statements One is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): Product of Pressure p and time t has the same dimension as that of coefficient of viscosity.

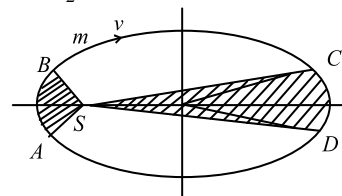
Reason (R): Coefficient of viscosity

$$= \frac{\text{Force}}{\text{Velocity gradient}}$$

Choose the **correct** answer from the options given below:

(1) Both **A** and **R** true, and **R** is correct explanation of **A**.
(2) Both **A** and **R** are true but **R** is NOT the correct explanation of **A**.
(3) **A** is true but **R** is false.
(4) **A** is false but **R** is true.

42. The figure shows elliptical orbit of a planet m about the sun S . The shaded area SCD is twice the shaded area SAB . If t_1 is the time for the planet to move from C to D and t_2 is the time to move from A to B then:

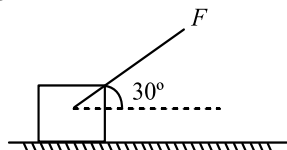


(1) $t_1 = 4t_2$ (2) $t_1 = 2t_2$
(3) $t_1 = t_2$ (4) $t_1 > t_2$

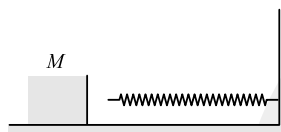
43. A clock has a continuously moving second's hand of 0.1 m length. The average acceleration of the tip of the hand (in ms^{-2}) is of the order of:

(1) 10^{-3} (2) 10^{-4}
(3) 10^{-2} (4) 10^{-1}

44. As shown in the figure a block of mass 10 kg lying on a horizontal surface is pulled by a force F acting at an angle 30° with horizontal. For $\mu_s = 0.25$, the block will just start to move for the value of F . (Given, $g = 10 \text{ ms}^{-2}$)



- (1) 25.2 N (2) 33.3 N
(3) 20 N (4) 35.7 N
45. The block of mass M moving on the frictionless horizontal surface collides with the spring of spring constant k and compresses it by length L . The maximum momentum of the block after collision is;



- (1) $L\sqrt{Mk}$ (2) $\frac{kL^2}{2M}$
(3) zero (4) $\frac{ML^2}{k}$

Chemistry

46. During photoelectric effect, threshold frequency:
- (1) increases as intensity of light increases.
 - (2) decreases as intensity of light increases.
 - (3) increases as wavelength of light increases.
 - (4) does not increase or decrease with intensity or wavelength of light used.
47. The total number of neutrons present in 36 g of water is:
- (1) $24 N_A$ (2) $32 N_A$
(3) $16 N_A$ (4) $20 N_A$
48. Minimum negative electron gain enthalpy among the following is of:
- (1) Se (2) O
(3) Te (4) S

49. The repulsive interaction of electron pair decreases in the order:
- (1) bond pair – bond pair > lone pair – bond pair > lone pair – lone pair.
 - (2) lone pair – lone pair > lone pair – bond pair > bond pair – bond pair.
 - (3) lone pair – lone pair > bond pair – bond pair > lone pair – bond pair.
 - (4) bond pair – bond pair > lone pair – lone pair > lone pair – bond pair.
50. A basic buffer solution contains equal concentration of base and its salt. If the value of K_b is 10^{-5} , then the pH of the buffer solution would be:
- (1) 11 (2) 10
(3) 12 (4) 9
51. A gas expands by 0.8 litre against a constant pressure of 1 atm. The work done in joule is: (Given: 1 L atm = 101.325 J)
- (1) +43.07 J
(2) -81.06 J
(3) +81.06 J
(4) -43.07 J
52. In a hydrogen spectrum, if transition of electron takes place from 6th to 3rd orbit, then the total number of spectral lines obtained in the Balmer series is:
- (1) 5 (2) 0
(3) 3 (4) 4
53. The electronic configuration of the element with maximum non-metallic character among the following is:
- (1) $1s^2 2s^2 2p^6 3s^2 3p^3$
(2) $1s^2 2s^2 2p^6 3s^2 3p^2$
(3) $1s^2 2s^2 2p^6 3s^2 3p^5$
(4) $1s^2 2s^2 2p^6 3s^2 3p^1$

54. Match List-I with List-II:

List-I (Molecule/ion)	List-II (Bond angle)
(A) SO_3	(I) 180°
(B) C_2H_2	(II) 109.5°
(C) SF_6	(III) 120°
(D) NH_4^+	(IV) 90° and 180°

Choose the **correct** answer from the options given below:

- (1) A-III, B-I, C-IV, D-II
 (2) A-II, B-III, C-IV, D-I
 (3) A-II, B-I, C-III, D-IV
 (4) A-III, B-I, C-II, D-IV

55. The number of significant figures present in 151 cm, 0.151 cm and 0.0151 cm respectively are:

- (1) 3, 3 and 3 (2) 3, 4 and 5
 (3) 3, 3 and 5 (4) 3, 4 and 4

56. Match List-I with List-II:

List-I (Ion)	List-II (No. of lone pair on central atom)
(A) NO_2^+	(I) 0
(B) I_3^-	(II) 1
(C) I_3^+	(III) 3
(D) NO_2^-	(IV) 2

Choose the **correct** answer from the options given below:

- (1) A-I, B-III, C-IV, D-II
 (2) A-II, B-III, C-IV, D-I
 (3) A-II, B-I, C-III, D-IV
 (4) A-III, B-I, C-II, D-IV

57. The incorrect postulates of the Dalton's atomic theory are:

- A. Atoms of different elements differ in mass.
 B. Matter consists of divisible atoms.
 C. Compounds are formed when atoms of different element combine in a fixed ratio.

D. All the atoms of given element have different properties including mass.

E. Chemical reactions involve reorganisation of atoms.

Choose the correct answer from the options given below:

- (1) (B), (D), (E) only (2) (A), (B), (D) only
 (3) (C), (D), (E) only (4) (B), (D) only

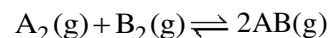
58. When 2.24 L of $\text{H}_2(\text{g})$ is mixed with 0.112 L of $\text{Cl}_2(\text{g})$ each at STP, the moles of $\text{HCl}(\text{g})$ formed is equal to:

- (1) 0.05 moles of $\text{HCl}(\text{g})$
 (2) 0.2 moles of $\text{HCl}(\text{g})$
 (3) 0.01 moles of $\text{HCl}(\text{g})$
 (4) 0.1 moles of $\text{HCl}(\text{g})$

59. Which of the following statement is **incorrect** regarding canal rays?

- (1) Canal rays carry negatively charged particles.
 (2) The behaviour of canal rays in the magnetic or electric field is opposite to that observed for electron or cathode rays.
 (3) Canal rays are also known as anode rays.
 (4) The characteristics of canal rays depends upon the material of electrodes and the nature of gas present in the cathode ray tube.

60. Determine the value of equilibrium constant (K_C) for the reaction:



If 5 moles of A_2 , 4 moles of B_2 and 6 moles of AB are placed in a 2 litre vessel and allowed to come to equilibrium. The final concentration of AB is 5 M:

- (1) 16.66 (2) 1.66
 (3) 166 (4) 0.1666

61. The heat of combustion of carbon to CO_2 is $-3.935 \times 10^5 \text{ J/mol}$. The heat released upon the formation of 88 g CO_2 from carbon and oxygen gas is:

- (1) -587 J/mol (2) -787 J/mol
 (3) -587 kJ/mol (4) -787 kJ/mol

62. Among the following the set of parameters that represents path function is:

- A. $q + w$ B. q
C. w D. $H - TS$

- (1) (B) and (C)
(2) (B), (C) and (D)
(3) (A), (B) and (C)
(4) (A) and (D)

63. Which of the following statement (s) is/are correct?

- A. At equilibrium the pressure exerted by the water molecules at a given temperature remains constant and is called the equilibrium vapour pressure of water.
B. Vapour pressure of water increases with increase in temperature.
C. Volatile liquids have very low vapour pressure than other liquid at the same temperature.
D. More volatile liquid has lower boiling point.

- (1) A, B and D (2) A and D
(3) A, B and C (4) Only C

64. **Assertion (A):** The ratio by volume of $\text{NH}_3 : \text{HCl} : \text{NH}_4\text{Cl}$ is 1 : 1 : 1 in the reaction
 $\text{NH}_3(\text{g}) + \text{HCl}(\text{g}) \rightarrow \text{NH}_4\text{Cl}(\text{s})$

Reason (R): Gay Lussac's law deals with gaseous reactants and products only.

- (1) Both **A** and **R** true, and **R** is correct explanation of **A**.
(2) Both **A** and **R** are true but **R** is NOT the correct explanation of **A**.
(3) **A** is true but **R** is false.
(4) **A** is false but **R** is true.

65. The $\frac{K_P}{K_C}$ ratio will be lowest in case of:

- (1) $\text{CO}(\text{g}) + \frac{1}{2}\text{O}_2(\text{g}) \rightleftharpoons \text{CO}_2(\text{g})$
(2) $\text{PCl}_5(\text{g}) \rightleftharpoons \text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g})$
(3) $7\text{H}_2(\text{g}) + 2\text{NO}(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g}) + 4\text{H}_2\text{O}(\text{g})$
(4) $\text{H}_2(\text{g}) + 2\text{NO}(\text{g}) \rightleftharpoons 2\text{HI}(\text{g})$

66. Match **List-I** with **List-II**:

List-I (Orbitals)	List-II (No. of angular nodes)
(A) 3s	(I) 0
(B) 4p	(II) 1
(C) 3d	(III) 3
(D) 5f	(IV) 2

Choose the **correct** answer from the options given below:

- (1) A-I, B-II, C-IV, D-III
(2) A-II, B-I, C-III, D-IV
(3) A-II, B-I, C-IV, D-III
(4) A-IV, B-I, C-II, D-III

67. Given below are two statements:

Statement-I: In cyclic process, ΔU and ΔH is zero.

Statement-II: When initial and final state of a system in a process are the same, the process is called a cyclic process.

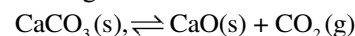
In the light of the above statements, choose the *most appropriate* answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

68. Which of the following ion has bond order equals to N_2 ?

- (1) O_2^{2+} (2) O_2^-
(3) O_2^{2-} (4) O_2^+

69. How much amount of CaCO_3 having percentage purity 25% produces 1.12 L of CO_2 at STP on heating?



- (1) 45 g (2) 10 g
(3) 25 g (4) 20 g

70. The **correct** order of the size of the following ions is:

- (1) $\text{Cl}^- > \text{K}^+ > \text{Ca}^{2+}$ (2) $\text{Ca}^{2+} > \text{K}^+ > \text{Cl}^-$
(3) $\text{Ca}^{2+} > \text{Cl}^- > \text{K}^+$ (4) $\text{K}^+ > \text{Cl}^- > \text{Ca}^{2+}$

71. Ultraviolet light of 8.4 eV falls on cesium surface (work function = 1.4 eV). The kinetic energy (in eV) of the fastest electron emitted is:
 (1) 9 eV (2) 7 eV
 (3) 3 eV (4) 2 eV
72. The **correct** increasing order of energy for the molecular orbital diagram of F_2 is:
 (1) $\sigma 1s < \sigma^* 1s < \sigma 2s < \sigma^* 2s < (\pi 2p_x = \pi 2p_y) < \sigma 2p_z < (\pi^* 2p_x = \pi^* 2p_y) < \sigma^* 2p_z$
 (2) $\sigma 1s < \sigma^* 1s < \sigma 2s < \sigma^* 2s < \sigma 2p_z < (\pi 2p_x = \pi 2p_y) < (\pi^* 2p_x = \pi^* 2p_y) < \sigma^* 2p_z$
 (3) $\sigma 1s < \sigma^* 1s < \sigma 2s < \sigma^* 2s < (\pi 2p_x = \pi 2p_y) < \sigma 2p_z < \sigma^* 2p_z < (\pi^* 2p_x = \pi^* 2p_y)$
 (4) $\sigma 1s < \sigma^* 1s < \sigma 2s < \sigma^* 2s < \sigma 2p_z < (\pi 2p_x = \pi 2p_y) < \sigma^* 2p_z < (\pi^* 2p_x = \pi^* 2p_y)$
73. What is the maximum number of orbitals that can be identified with the following quantum numbers?
 $n = 4, l = +2, m = 0$
 (1) 1 (2) 2
 (3) 3 (4) 4
74. The amount of heat that is required to raise the temperature of 108 g of aluminium from 15°C to 35°C is:
 [Given: Molar heat capacity of Al is 24 J/mol K]
 (1) 0.86 kJ (2) 1.92 kJ
 (3) 1.07 kJ (4) 2.76 kJ
75. If an ion contains 8 protons, 8 neutrons and 10 electrons, then the charge present on it will be:
 (1) -1 (2) -2
 (3) +1 (4) +2
76. The **correct** increasing order of the covalent character in the following molecule is:
 (1) $\text{AlCl}_3 < \text{MgCl}_2 < \text{NaCl}$
 (2) $\text{MgCl}_2 < \text{NaCl} < \text{AlCl}_3$
 (3) $\text{NaCl} < \text{MgCl}_2 < \text{AlCl}_3$
 (4) $\text{MgCl}_2 < \text{AlCl}_3 < \text{NaCl}$

77. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion (A): SF_6 molecule is octahedral.
Reason (R): The value of dipole moment of SF_6 molecule is zero.
 In the light of the above statements, choose the **correct** answer from the options given below:
 (1) A is true but R is false.
 (2) A is false but R is true.
 (3) Both A and R are true and R is the correct explanation of A.
 (4) Both A and R are true but R is NOT the correct explanation of A.
78. Which of the following methods of expressing the concentration of a solution are not temperature dependent?
 A. Molarity B. Mole Fraction
 C. Molality D. Weight percent
 (1) (A), (B) and (C) (2) (B), (C) and (D)
 (3) (A) and (C) (4) (A) and (D)
79. Given below are two statements:
Statement-I: K_w of water decreases with increase in temperature.
Statement-II: Conjugate acid of NH_3 is NH_4^+ .
 In the light of the above statements, choose the *most appropriate* answer from the options given below:
 (1) Statement I is correct but Statement II is incorrect.
 (2) Statement I is incorrect but Statement II is correct.
 (3) Both Statement I and Statement II are correct.
 (4) Both Statement I and Statement II are incorrect.
80. For the given reaction, the equilibrium constant is 25.
 $\text{N}_2 + 3\text{H}_2 \rightleftharpoons 2\text{NH}_3$
 Calculate value of equilibrium constant for the following reaction.
 $\text{NH}_3 \rightleftharpoons \frac{1}{2}\text{N}_2 + \frac{3}{2}\text{H}_2$
 (1) 0.25 (2) 0.4
 (3) 0.2 (4) 0.625

81. Match List-I with List-II:

List-I (Element)	List-II (Group number)
(A) N	(I) 2
(B) Be	(II) 3
(C) Sc	(III) 17
(D) Cl	(IV) 15

Choose the **correct** answer from the options given below:

- (1) A-I, B-II, C-IV, D-III
 (2) A-II, B-I, C-III, D-IV
 (3) A-II, B-I, C-IV, D-III
 (4) A-IV, B-I, C-II, D-III

82. Intensive property among the following is:

- (1) Surface area (2) Electrode potential
 (3) Gibbs energy (4) Heat capacity

83. The set of quantum numbers which is **not** possible for an electron is:

- (1) $n = 4, l = 1, m = 0, s = -\frac{1}{2}$
 (2) $n = 3, l = 2, m = +1, s = +\frac{1}{2}$
 (3) $n = 3, l = 1, m = -2, s = -\frac{1}{2}$
 (4) $n = 4, l = 0, m = 0, s = +\frac{1}{2}$

84. Pure PCl_5 is introduced into an evacuated chamber and comes to equilibrium at 247°C and 2.0 atm. The equilibrium gaseous mixture contains 40% chlorine by volume. Calculate K_p at 247°C for the reaction:

- (1) 0.4 atm (2) 40 atm
 (3) 1.6 atm (4) 0.16 atm

85. What is the volume occupied by 5.2 g of a gas at STP?

[Given: Vapour density of gas is 11.2]

- (1) 5.2 L (2) 2.24 L
 (3) 22.4 L (4) 1.12 L

86. According to Mosely, a straight-line graph is obtained on plotting:

- (1) The frequencies of characteristic X-rays of elements against their atomic numbers.
 (2) The square of the frequencies of characteristic X-rays of elements against their atomic numbers.
 (3) The square root of the frequencies of characteristic X-rays of elements against their atomic number.
 (4) The reciprocal of the frequencies of characteristic X-rays of elements against their atomic numbers.

87. Which block of the periodic table contains elements with the general outer electronic configuration?

$$(n-2)f^{1-14} (n-1)d^{0-1} ns^2$$

- (1) s-block (2) p-block
 (3) d-block (4) f-block

88. Identify the **correct** statements among the following:

- A. Atomic radius of the elements decreases as one moves down the first group of the periodic table.
 B. Mendeleev left a gap under aluminium and silicon and called these elements Eka-aluminium and Eka-silicon.
 C. Unnilunium is the IUPAC name of the element with atomic number 101.
 (1) A and B only (2) B and C only
 (3) A and C only (4) A, B and C

89. Match List-I with List-II:

List-I (Compound)	List-II (Shape)
(A) Cl_2O	(I) Trigonal pyramidal
(B) BrF_5	(II) Bent
(C) XeO_3	(III) Linear
(D) XeF_2	(IV) Square pyramidal

Choose the **correct** answer from the options given below:

- (1) A-II, B-IV, C-I, D-III
 (2) A-II, B-III, C-IV, D-I
 (3) A-II, B-I, C-III, D-IV
 (4) A-III, B-I, C-II, D-IV

90. Radius of 1st Bohr's orbit of Be^{3+} ion is:

- (1) 1.592 Å (2) 0.132 Å
 (3) 0.364 Å (4) 1.894 Å

Botany

91. All the universal rules of ICBN are **correct** among the following **except**:

- (1) handwritten scientific names should be separately underlined.
- (2) every species should have a generic name and a specific epithet.
- (3) scientific names are generally in Latin.
- (4) generic and specific names should be written starting with small letters.

92. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R):

Assertion (A): Viroids are found to be a free RNA.

Reason (R): Viroids lacked the protein coat that is found in viruses.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) **A** is true but **R** is false.
- (2) **A** is false but **R** is true.
- (3) Both **A** and **R** are true and **R** is the correct explanation of **A**.
- (4) Both **A** and **R** are true but **R** is NOT the correct explanation of **A**.

93. Fusion between one large static female gamete and smaller motile male gamete is termed as ____ as seen in ____.

- (1) oogamous, *Volvox*
- (2) isogamous, *Spirogyra*
- (3) anisogamous, *Fucus*
- (4) oogamous, *Ulothrix*

94. Match **List-I** with **List-II**.

List-I	List-II
(A) Chemotaxonomy (I)	uses chemical constituents of plants to resolve confusions
(B) Numerical taxonomy (II)	based on chromosome number
(C) Cytotaxonomy (III)	each character is given equal importance

Choose the **correct** answer from the options given below:

- (1) A-I, B-II, C-III
- (2) A-I, B-III, C-II
- (3) A-III, B-I, C-II
- (4) A-II, B-III, C-I

95. Which of the following statements is **correct**?

- (1) Asexual reproduction in fungi is by oospores, ascospores and basidiospores.
- (2) Fusion of two nuclei between two motile or non-motile gametes called plasmogamy.
- (3) Fusion of two protoplasm is called karyogamy.
- (4) Meiosis in zygote results in the formation of haploid spores during sexual cycle of fungi.

96. Which of the following plants has an irregular flower?

- (1) Canna
- (2) Bean
- (3) Pea
- (4) Mustard

97. In a monocot seed, the outer covering of endosperm separates the embryo by a proteinaceous layer called:

- (1) scutellum.
- (2) aleurone.
- (3) coleoptile.
- (4) coleorhiza.

98. Match **List-I** with **List-II**.

List-I	List-II
(A) <i>domestica</i>	(I) Family
(B) Diptera	(II) Phylum
(C) <i>Musca</i>	(III) Order
(D) Arthropoda	(IV) Species
(E) Muscidae	(V) Genus

Choose the **correct** answer from the options given below:

- (1) A-V, B-IV, C-II, D-I, E-III
- (2) A-IV, B-III, C-V, D-II, E-I
- (3) A-IV, B-III, C-V, D-I, E-II
- (4) A-IV, B-III, C-II, D-V, E-I

99. Flagellated pyriform (pear-shaped) gametes are characteristic of:

- (1) chlorophyceae.
- (2) rhodophyceae.
- (3) cyanobacteria.
- (4) phaeophyceae.

- 100.** Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Pteridophytes show events which are precursor to seed habit.

Reason R: The development of the zygotes into young embryos take place within the female gametophytes which remain retained on the sporophyte.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is NOT the correct explanation of A.

- 101.** Given below are two statements:

Statement I: The form of food stored in red algae is structurally similar to glycogen and amylopectin.

Statement II: Majority of the red algae are marine, even present at great depths in oceans where relatively little light penetrates.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

- 102.** Read the following statements and choose the **correct** ones.

- A. Like bryophytes and pteridophytes, in gymnosperms the male and the female gametophytes have an independent free-living existence.
- B. In the predominant gametophytic stage of the life cycle of a moss, the first stage is the protonema stage, which develops directly from a spore.

- C. In the predominant gametophytic stage of the life cycle of a moss, the second stage is the leafy stage, which develops from the secondary protonema as a lateral bud.

- (1) A only
- (2) A and B
- (3) B and C
- (4) A and C

- 103.** The major pigments in the class of algae that store their food in the form of floridean starch are:

- (1) chlorophyll a, c.
- (2) chlorophyll a, c, phycoerythrin.
- (3) chlorophyll a, b, fucoxanthin.
- (4) chlorophyll a, d, phycoerythrin.

- 104.** The type of placentation where ovary is one chambered but it becomes two-chambered due to the formation of the false septum is found in:

- (1) *Primrose*.
- (2) *Dianthus*.
- (3) *Argemone*.
- (4) china rose.

- 105.** Pick the **incorrectly** matched pair.

- (1) Stilt roots: provide support.
- (2) Tap roots of sweet potato: store food.
- (3) Underground stem of ginger: act as organs of perennation.
- (4) Thorns of *Citrus*: modified axillary buds for protection.

- 106.** Given below are two statements:

Statement I: In racemose type of inflorescences, the growth of the main axis of the plant is not limited.

Statement II: In racemose type of inflorescences the flowers borne in acropetal succession.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

- 107.** Identify the **false** statements from the following.
- No non-living object exhibits metabolism.
 - Brain-dead human patients have selfconsciousness.
 - Each rank or taxon represents a unit of nomenclature.
 - Based on characteristics, all living organisms can be classified into various taxa.

Choose the *most appropriate* answer from the options given below:

- (1) A and B (2) B and C
- (3) A and C (4) C and D

- 108.** Given below are two statements:

Statement I: Archaeobacteria live in some of 'most harsh habitat conditions.'

Statement II: The archaeobacteria and other bacteria have same type of cell wall structure.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

- 109.** A gymnosperm which is unbranched, dioecious and possesses coralloid roots, is:

- (1) *Pinus*. (2) *Cycas*.
- (3) *Sequoia*. (4) *Cedrus*.

- 110.** Match **List-I** with **List-II**.

List-I	List-II
(A) Diatoms	(I) Plasmodium
(B) Euglenoids	(II) Mixotrophic nutrition
(C) Dinoflagellates	(III) Red tide
(D) Slime moulds	(IV) Oceanic chief producers

Choose the **correct** answer from the options given below:

- (1) A-IV, B-II, C-III, D-I
- (2) A-I, B-II, C-III, D-IV
- (3) A-IV, B-I, C-II, D-III
- (4) A-III, B-II, C-I, D-IV

- 111.** Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion (A): Viruses contain genetic material.
Reason (R): No virus contains both RNA and DNA.
 In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is NOT the correct explanation of A.

- 112.** The ovary demonstrated in the given figure is of which type?



- (1) Superior (2) Inferior
- (3) Half-inferior (4) None of these

- 113.** Choose the **incorrect** statement from the following w.r.t. casparian strips in dicotyledonous roots.

- (1) Waxy in nature and made up of suberin.
- (2) Obstruct the flow of water in the roots.
- (3) Present in the innermost layer of cortex.
- (4) It is a water-impermeable deposition in pericycle.

- 114.** In mango, the pericarp is well differentiated into an outer thin 'A', a middle fleshy edible 'B' and an inner stony hard 'C'. B, C and A, respectively, are:

- (1) epicarp, mesocarp, endocarp.
- (2) epicarp, endocarp, mesocarp.
- (3) mesocarp, endocarp, epicarp.
- (4) endocarp, epicarp, mesocarp.

- 115.** Given below are two statements:

Statement I: In isobilateral leaf, the stomata are present only on the upper epidermal surface.

Statement II: In isobilateral leaf, the mesophyll is well differentiated into palisade and spongy parenchyma.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

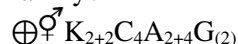
116. Which of the following shows imbricate aestivation?

- (a) *Calotropis* (b) China rose
(c) Lady's finger (d) Cotton
(e) *Cassia* (f) Gulmohur
(1) (a), (b) and (d) (2) (c) and (e)
(3) (e) and (f) (4) (d), (e) and (f)

117. The innermost layer of the cortex in dicot root is called:

- (1) endodermis. (2) pericycle.
(3) epiblema. (4) conjunctive tissue.

118. The floral formula given below represents which family?



- (1) Compositae (2) Cruciferae
(3) Malvaceae (4) Leguminosae

119. Given below are two statements:

Statement I: Vascular bundles are not a part of stele.

Statement II: All tissues inside the endodermis constitute the stele.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

120. Sac-fungi includes all the following, **except**:

- (1) *Penicillium*. (2) *Saccharomyces*.
(3) *Agaricus*. (4) truffles.

121. Choose the correct sequence of components of dicot stem in the sequential order (inner to outer):

- (1) Phloem, pericycle, endodermis, parenchyma, collenchyma, epidermis
(2) Phloem, cambium, metaxylem, protoxylem, pith
(3) Epidermis, parenchyma, collenchyma, hypodermis, pericycle
(4) Epidermis, hypodermis, vascular bundle, ground tissue

122. All the following are **correct** w.r.t. dorsiventral leaf, **except**:

- (1) mesophyll, which possesses chloroplasts and carry out photosynthesis, is made up of sclerenchyma.
(2) the abaxial epidermis generally bears more stomata than the adaxial epidermis.
(3) mesophyll has two types of cells – the palisade parenchyma and the spongy parenchyma.
(4) both adaxial and abaxial epidermis of the leaf has a conspicuous cuticle.

123. Match **List-I** with **List-II**.

List-I	List-II
(A) Lily	(I) Perigynous Flower
(B) Plum, Peach, Rose	(II) Zygomorphic flower
(C) Pea, Bean, Gulmohur	(III) Syncarpous Ovary
(D) Mustard, Tomato	(IV) Epiphyllous stamens

Choose the **correct** answer from the options given below:

- (1) A-IV, B-I, C-II, D-III
(2) A-IV, B-I, C-III, D-II
(3) A-II, B-I, C-IV, D-III
(4) A-III, B-II, C-I, D-IV

124. Match **List-I** with **List-II**.

List-I	List-II
(A) Parasitic fungus on mustard	(I) <i>Puccinia</i>
(B) Feed on dead substrates	(II) <i>Neurospora</i>
(C) Wheat rust pathogen	(III) Saprophytes
(D) Used for biochemical and genetic work	(IV) <i>Albugo</i>

Choose the **correct** answer from the options given below:

- (1) A-I, B-II, C-III, D-IV
(2) A-IV, B-III, C-I, D-II
(3) A-IV, B-III, C-II, D-I
(4) A-II, B-I, C-IV, D-III

125. Match List-I with List-II.

List-I	List-II
(A) <i>Calotropis</i>	(I) Sepals or petals in a whorl just touch one another at the margin, without overlapping.
(B) <i>Cassia</i>	(II) The margins of sepals or petals overlap one another but not in any particular direction.
(C) Bean	(III) Five petals, the largest (standard) overlaps the two lateral petals (wings) which in turn overlap the two smallest anterior petals (Keel).

Choose the **correct** answer from the options given below:

- (1) A-III, B-I, C-II
- (2) A-II, B-I, C-III
- (3) A-I, B-II, C-III
- (4) A-III, B-II, C-I

126. Which pair of the following statements is incorrect?

- A. When a flower can be divided into two equal radial halves in any radial plane passing through the centre, it is said to be actinomorphic, eg. pea.
- B. Bean, gram and pea have non-endospermous seeds.
- C. The hilum is a scar on the seed coat through which the developing seeds were attached to the fruit.
- D. Maize has two cotyledons.

Choose the *most appropriate* answer from the options given below:

- (1) B and C
- (2) A and C
- (3) B and D
- (4) A and D

127. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion (A): The parenchymatous cells which lie between the xylem and the phloem are called pericycle.

Reason (R): Initiation of lateral roots and vascular cambium during secondary growth takes place in pericycle.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is NOT the correct explanation of A.

128. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: The 'ring' arrangement of vascular bundles is a characteristic of dicot stem.

Reason R: Each vascular bundle is conjoint, open and with endarch protoxylem in dicot stem. In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is NOT the correct explanation of A.

129. Match List-I with List-II.

List-I	List-II
(A) Swollen leaf base	(I) Pulvinus
(B) Green expanded part of leaf	(II) Root cap
(C) Thimble like structure	(III) Lamina
(D) Attaches the leaf to stem	(IV) Leaf base

Choose the **correct** answer from the options given below:

- (1) A-II, B-III, C-I, D-IV
- (2) A-III, B-II, C-I, D-IV
- (3) A-I, B-III, C-II, D-IV
- (4) A-II, B-I, C-III, D-IV

130. Match List-I with List-II.

List-I	List-II
(A) Prevents the loss of water	(I) Cuticle
(B) Enclose stomatal pore	(II) Subsidiary cells
(C) Regulate the process of transpiration and gaseous exchange	(III) Guard cells
(D) Epidermal cells in vicinity of guard cells specialized in their shape and size	(IV) Stomata

Choose the **correct** answer from the options given below:

- (1) A-II, B-I, C-III, D-IV
- (2) A-I, B-III, C-II, D-IV
- (3) A-II, B-IV, C-III, D-I
- (4) A-I, B-III, C-IV, D-II

131. Which of the following statements about *Triticum aestivum* are correct?

- A. The plant has an adventitious root system with stilt roots.
- B. Leaves are simple, alternate, with parallel venation, a sheathing leaf base, and a long ligule.
- C. The male flowers have three stamens and a rudimentary gynoecium.
- D. The female flowers in the plant are characterised by a tricarpeal, syncarpous gynoecium.
- E. The stem is herbaceous, erect, cylindrical, and fistular with distinct nodes and internodes.

Choose the correct answer:

- (1) A, B, and E only
- (2) B, C, and E only
- (3) A, C, and E only
- (4) A, D, and E only

132. Match List-I with List-II.

List-I (Diagrammatic representation)	List-II (Placentation type)
(A) 	(I) Axile
(B) 	(II) Parietal
(C) 	(III) Free central
(D) 	(IV) Basal

Choose the **correct** answer from the options given below:

- (1) A-II, B-III, C-I, D-IV
- (2) A-III, B-IV, C-I, D-II
- (3) A-I, B-IV, C-II, D-III
- (4) A-III, B-I, C-II, D-IV

133. Choose the **incorrect statement from the following w.r.t. trichomes.**

- A. They are usually multicellular.
- B. They may be branched or unbranched.
- C. They help in preventing water loss due to transpiration.
- D. They are always unicellular.

Choose the **most appropriate** answer from the options given below:

- (1) A and B
- (2) C only
- (3) B and D
- (4) D only

134. Match List-I with List-II.

List-I	List-II
(A) Enclosed by coleoptile	(I) Aleurone layer
(B) Stores food	(II) Plumule
(C) Outer covering of endosperm in monocot seed	(III) Radicle
(D) Enclosed by coleorhiza	(IV) Endosperm

Choose the **correct** answer from the options given below:

- (1) A-II, B-IV, C-I, D-III
- (2) A-III, B-II, C-I, D-IV
- (3) A-I, B-IV, C-II, D-III
- (4) A-II, B-I, C-III, D-IV

- 135.** Identify the **incorrect** statement w.r.t stomata.
- (1) Stomata are present in the epidermis of leaves.
 - (2) Each stoma is composed of two bean-shaped guard cells.
 - (3) The inner wall of guard cells is thin.
 - (4) The guard cells possess chloroplasts and regulate the opening and closing of stomata.

Zoology

- 136.** Excretory organ of hemichordates is:
- (1) green gland. (2) flame cells.
 - (3) nephridia. (4) proboscis gland.
- 137.** Given below are some statements (A-D).
- A. In male cockroaches, a mushroom-shaped gland is present in the 6th-7th abdominal segments.
 - B. Cockroaches are uricotelic.
 - C. The urecose glands help in respiration.
 - D. Blood from the sinuses enters the heart through the ostia and is pumped anteriorly to the sinuses again.
- Choose the **correct** statements from the options given below:
- (1) A, B and D only
 - (2) B and C only
 - (3) A and D only
 - (4) B and D only
- 138.** Which of the following are considered as macromolecules in the acid-insoluble fraction?
- (a) Proteins
 - (b) Nucleic acids
 - (c) Lipids
 - (d) Polysaccharides
 - (e) Amino acids
- Choose the correct answer from the options given below:
- (1) (a), (b), and (d) only
 - (2) (c), (d), and (e) only
 - (3) (a), (c), and (e) only
 - (4) (b), (c), and (d) only

- 139.** Match **List-I** with **List-II**:

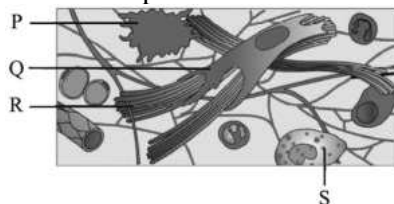
List-I	List-II
(A) <i>Myxine</i>	(I) Sawfish
(B) <i>Pristis</i>	(II) Flying fish
(C) <i>Exocoetus</i>	(III) Fighting fish
(D) <i>Betta</i>	(IV) Hagfish

Choose the **correct** answer from the options given below:

- (1) A-II, B-I, C-IV, D-III
 - (2) A-I, B-III, C-II, D-IV
 - (3) A-IV, B-I, C-II, D-III
 - (4) A-III, B-II, C-IV, D-I
- 140.** Which of the following is **not** secreted by the exocrine gland?
- (1) Saliva (2) Ear wax
 - (3) Insulin (4) Mucus
- 141.** Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R):
Assertion (A): Lipids come under acid-soluble fraction.
Reason (R): Lipids have a small molecular weight as compared to biomacromolecules.
 In the light of the above statements, choose the **correct** answer from the options given below:
- (1) A is true but R is false.
 - (2) A is false but R is true.
 - (3) Both A and R are true and R is the correct explanation of A.
 - (4) Both A and R are true but R is NOT the correct explanation of A.

- 142.** Given below are two statements:
- Statement I:** Cyclostomes have a sucking and circular mouth without jaws.
- Statement II:** Hemichordates have a rudimentary structure in the trunk region called a stomochord.
- In the light of the above statements, choose the *most appropriate* answer from the options given below:
- (1) Statement I is correct but Statement II is incorrect.
 - (2) Statement I is incorrect but Statement II is correct.
 - (3) Both Statement I and Statement II are correct.
 - (4) Both Statement I and Statement II are incorrect.

143. Select the option with correct labelling.



- (1) P – Fibroblast, R – Collagen fibres
- (2) R – Collagen fibres, S – Macrophage
- (3) Q – Fibroblast, S – Mast cell
- (4) P – Macrophage, S – Fibroblast

144. Given below are some statements (A-D).

- A. GLUT-4 enables glucose transport into the cells.
- B. Water is the most abundant chemical in the living organism.
- C. Paper and cotton fibre are cellulosic in nature.
- D. Chitin is a monosaccharide.

Choose the **correct** statements from the options given below:

- (1) A, B and D only
- (2) B and D only
- (3) A and C only
- (4) A, B and C only

145. *Chaetopleura* (Chiton) belongs to which of the following phyla?

- (1) Echinodermata
- (2) Mollusca
- (3) Arthropoda
- (4) Annelida

146. Match List-I with List-II:

List-I	List-II
(A) Loose connective tissue	(I) Tendons
(B) Dense regular connective tissue	(II) Adipose tissue
(C) Dense irregular connective tissue	(III) Cartilage tissue
(D) Specialised connective tissue	(IV) Skin

Choose the **correct** answer from the options given below:

- (1) A-II, B-I, C-IV, D-III
- (2) A-I, B-II, C-III, D-IV
- (3) A-IV, B-II, C-III, D-I
- (4) A-III, B-II, C-IV, D-I

147. Malonate acts as a competitive inhibitor for succinic dehydrogenase because it closely resembles the substrate _____ in structure.

- (1) succinate
- (2) niacin
- (3) carboxypeptidase
- (4) adenine

148. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: Cnidarians are diploblastic.

Reason R: In diploblastic organisms, cells are arranged in two embryonic layers, an external endoderm and an inner ectoderm.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is NOT the correct explanation of A.

149. Smooth muscles are:

- (1) involuntary, cylindrical and striated.
- (2) voluntary, spindle-shaped and uninucleate.
- (3) involuntary, fusiform and non-striated.
- (4) voluntary, multinucleated and cylindrical.

150. Given below are two statements:

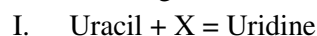
Statement I: Ctenophores are exclusively marine and diploblastic organisms.

Statement II: Bioluminescence is well-marked in ctenophores.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

151. Refer to the given reaction.



What do X and Y represent here?

- (1) X- Phosphate group, Y- Sugar molecules
- (2) X- Sugar molecules, Y- Phosphate group
- (3) X- Sugar molecules, Y- Nitrogenous base
- (4) X- Nitrogenous base, Y- Sugar molecules

152. Given below are two statements:

Statement I: Epithelial tissues are specialised for linking and connecting other tissues/organs.

Statement II: Ligament is a type of dense irregular connective tissue.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

153. Which of the following organic compounds is the main constituent of lecithin?

- (1) Arachidonic acid
- (2) Phospholipid
- (3) Cholesterol
- (4) Phosphoprotein

154. Given below are some statements (A-D).

- A. Amphibians have internal fertilisation and direct development.
- B. The skin of amphibians is moist and without scales.
- C. The body of an amphibian is divided into a head and a trunk.
- D. Amphibians have three-chambered heart.

Choose the **correct** statements from the options given below:

- (1) A and D only
- (2) A and B only
- (3) B, C and D only
- (4) D only

155. Select the **incorrectly** matched pair.

- (1) Neurons-Nerve cells
- (2) Fibroblast-Areolar tissue
- (3) Osteocytes-Bone cells
- (4) Chondrocytes-Smooth muscle cells

156. Match **List-I** with **List-II**:

List-I	List-II
(A) The protein portion of the enzymes	(I) Ricin
(B) Possess peptide bonds	(II) Apoenzyme
(C) Cell wall material in plants	(III) Cellulose
(D) Secondary metabolite	(IV) Collagen

Choose the **correct** answer from the options given below:

- (1) A-II, B-III, C-I, D-IV
- (2) A-II, B-IV, C-III, D-I
- (3) A-III, B-I, C-IV, D-II
- (4) A-III, B- IV, C-I, D-II

157. Which of the following organisms is radially symmetrical but has a bilateral symmetry in larva?

- (1) *Ascaris* (Roundworm)
- (2) *Nereis* (Ringworm)
- (3) *Pila* (Apple snail)
- (4) *Asterias* (Starfish)

158. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Cell junctions are present in the epithelial tissues.

Reason R: Only one type of cell junction is found in animal tissues.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is NOT the correct explanation of A.

159. How many carbons (including carboxyl carbon) are found in palmitic acid?

- (1) 15
- (2) 16
- (3) 17
- (4) 18

160. Match **List-I** with **List-II**:

List-I	List-II
(A) Spongocoel	(I) Arthropoda
(B) Dorso-ventrally flattened body	(II) Cnidaria
(C) Coelenteron cavity	(III) Porifera
(D) Chitinous exoskeleton	(IV) Platyhelminthes

Choose the **correct** answer from the options given below:

- (1) A-III, B-IV, C-I, D-II
- (2) A-IV, B-III, C-II, D-I
- (3) A-III, B-IV, C-II, D-I
- (4) A-II, B- IV, C-III, D-I

- 161.** Intercalated discs at fusion points:
- allows the cells to contract as a unit.
 - allows the cells to contract separately.
 - contract one after another as a unit.
 - None of these
- 162.** Arrange the following steps in the correct order to describe the catalytic cycle of an enzyme:
- The enzyme releases the products of the reaction, becoming free to bind another substrate molecule.
 - The substrate binds to the enzyme's active site.
 - The enzyme changes its shape, fitting more tightly around the substrate.
 - The active site breaks the chemical bonds of the substrate, forming an enzyme-product complex.
- (b), (c), (d), (a)
 - (c), (b), (d), (a)
 - (d), (c), (b), (a)
 - (a), (d), (c), (b)
- 163.** Identify the **correct** option with respect to the figures (A and B) given below.

**A****Function****B****Location**

- | | | |
|--------------|---|----------------------------|
| (1) A | Protection against chemical and mechanical stress | Stomach |
| (2) B | Secrete mucus, saliva, earwax, oil, milk, digestive enzymes | Blood vessels |
| (3) A | Secretion and absorption | Proximal convoluted tubule |
| (4) B | Mainly present in the inner surface of hollow organs | Fallopian tube |

- 164.** In which one of the following organisms and its respiratory organs are **correctly** matched?
- Starfish - Pharyngeal Gills
 - King crab - Book gills
 - Cuttlefish - Book lungs
 - Earthworm - Tracheal system

- 165.** Given below are some statements (A-D).
- Glycerol is trihydroxy propane.
 - Lipids are called fats and oils based on their melting point.
 - Tertiary structure is necessary for many biological activities of proteins.
 - In glycogen, the left end is reducing and the right end is non-reducing.
- Choose the **incorrect** statement(s) from the options given below:
- A and D only
 - A and B only
 - B, C and D only
 - D only

- 166.** Identify the correct set of sensory organs in cockroaches.
- Maxillary palps
 - Labial palps
 - Phallomere
 - Anal styles
 - Anal cerci
- (a), (c), and (d) only
 - (a), (b), and (e) only
 - (b), (d), and (e) only
 - (c), (d), and (e) only

- 167.** Match **List-I** with **List-II**:

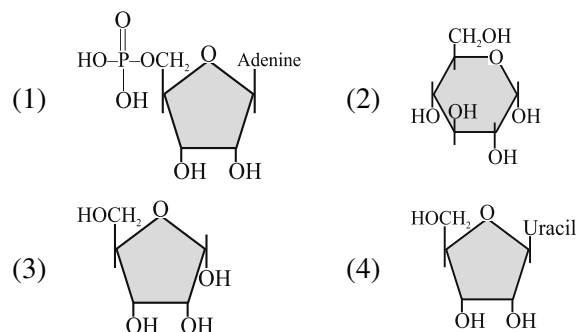
List-I**List-II**

- | | |
|-----------------|-------------------------|
| (A) Radula | (I) Respiratory organ |
| (B) Parapodia | (II) Helps in excretion |
| (C) Flame cells | (III) Helps in swimming |
| (D) Gills | (IV) Rasping organ |

Choose the **correct** answer from the options given below:

- A-IV, B-III, C-I, D-II
- A-IV, B-III, C-II, D-I
- A-III, B-IV, C-II, D-I
- A-II, B-IV, C-III, D-I

- 168.** Which of the following structures depicts a nucleoside?



- 169.** Given below are some statements (A-D).
- A. Bones have a hard and non-pliable ground substance.
 - B. The matrix of bone is rich in calcium salt and free from collagen fibres.
 - C. Bone marrow in some bones is the site of the production of blood cells.
 - D. The bone cells are called osteocytes.
- Choose the **correct** statements from the options given below:
- (1) A and B only (2) B and C only
(3) A, C and D only (4) A, B, C and D

- 170.** Match **List-I** with **List-II**:

List-I	List-II
(A) Vector	(I) <i>Limulus</i>
(B) Economically important insect	(II) <i>Locusta</i>
(C) Gregarious pest	(III) <i>Anopheles</i>
(D) Living fossil	(IV) <i>Bombyx</i>

Choose the **correct** answer from the options given below:

- (1) A-II, B-IV, C-III, D-I
(2) A-I, B-III, C-IV, D-II
(3) A-III, B-IV, C-II, D-I
(4) A-IV, B-I, C-III, D-II

- 171.** Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
- Assertion (A):** Cellulose is a homopolymer.
Reason (R): Cellulose is made of several glucose units.
- In the light of the above statements, choose the **correct** answer from the options given below:
- (1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is NOT the correct explanation of A.

- 172.** Given below are two statements:
- Statement I:** Neurons make up more than one-half of the volume of neural tissue.
Statement II: Tight junctions help to stop the leakage of substances across a tissue.
- In the light of the above statements, choose the *most appropriate* answer from the options given below:
- (1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

- 173.** Which of the following would be a characteristic of the animal shown in the given diagram?



- (1) It belongs to Cephalochordata.
(2) It migrates to freshwater for spawning.
(3) Exclusively marine.
(4) Gills are covered by operculum.

- 174.** Which of the following acids is used for chemical analysis of living tissues?
- (1) HCl (2) Cl_3CCOOH
(3) Acetic acid (4) H_2SO_4

- 175.** Match **List-I** with **List-II**:

List-I	List-II
(A) Camouflage	(I) Winter sleep
(B) Aestivation	(II) Ability to change colour to hide from their enemies
(C) Hibernation	(III) Summer sleep
(D) Poikilotherm	(IV) Cold-blooded

Choose the **correct** answer from the options given below:

- (1) A-II, B-III, C-I, D-IV
(2) A-III, B-II, C-I, D-IV
(3) A-I, B-II, C-III, D-IV
(4) A-II, B-I, C-III, D-IV

176. Given below are some statements (A-D).
- A. Polyps produce medusa sexually.
 - B. Metamerism is seen in earthworms.
 - C. Medusa forms polyps asexually.
 - D. Corals have a skeleton composed of calcium carbonate.

Choose the **correct** statements from the options given below:

- (1) A and D only (2) A and B only
(3) B, C and D only (4) B and D only

177. Given below are two statements:

Statement I: Inorganic compounds like sulphate, phosphate, etc., are seen in the acid-insoluble fraction.

Statement II: Arachidonic acid has 12 carbon atoms including the carboxyl carbon.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

178. Arrange the following events in the life cycle of Cyclostomata in the correct sequence:

- (a) Spawning
- (b) Migrate to freshwater
- (c) Death of parent cyclostome
- (d) Return of larvae to the ocean
- (e) Metamorphosis of larvae

- (1) (a), (b), (c), (d), (e) (2) (b), (a), (c), (e), (d)
(3) (a), (b), (d), (c), (e) (4) (e), (d), (c), (b), (a)

179. Given below are some statements (A-D).

- A. Acid insoluble fraction has only four types of organic compounds.
- B. Biomacromolecules have a molecular weight in the range of 10,000 Da and above.
- C. Molecular weights less than one thousand Dalton are usually referred to as macromolecules.
- D. All the carbon-containing compounds in a living tissue are called biomolecules.

Choose the **correct** statements from the options given below:

- (1) A and D only
(2) A and B only
(3) B and C only
(4) A, B and D only

180. Given below are two statements:

Statement I: An open circulatory system is found in most arthropods.

Statement II: In Aves, skin is dry without glands except the oil gland at the base of the tail.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

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CHEMISTRY

46. (4)
47. (3)
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BOTANY

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ZOOLOGY

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HINTS & SOLUTIONS

Physics

1. (4) [Class 11th, Ch. 4, Page 101]

$$T - f = m_1 a \quad \dots (1)$$

$$m_2 g - T = m_2 a \quad \dots (2)$$

2. (1) [Class 11th, Ch. 7, Page 134]

$$\Delta U = \frac{mgh}{1 + \frac{h}{R}} = \frac{1 \times 10 \times (3R)}{1 + 3} = \frac{30R}{4}$$

$$\Delta U = 48 \times 10^6 \text{ Joule}$$

3. (4) [Class 11th, Ch. 6, Page 103]

$\vec{A} \cdot \vec{B} = 0$, because vectors are perpendicular to each other.

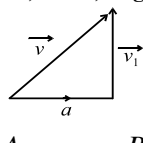
4. (3) [Class 11th, Ch. 6, Page 133]

Since both will remain always diametrically opposite so angular velocity will be same so time period of rotation will be same.

5. (3) [Class 11th, Ch. 6, Page 95]

$$X_{CM} = \frac{m_1(a) - \frac{m_1}{4}\left(\frac{3a}{2}\right)}{m_1 - \frac{m_1}{4}} = \frac{5a}{6}$$

6. (2) [Class 11th, Ch. 4, Page 64]



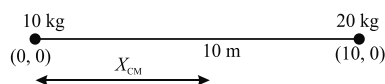
$$(vt)^2 = (v_1 t)^2 + a^2$$

$$t = \sqrt{\frac{a^2}{v^2 - v_1^2}}$$

7. (3) [Class 11th, Ch. 6, Page 106]

$$\begin{aligned} \vec{L} &= m(\vec{r}_{CM} \times \vec{V}_{CM}) \\ &= 2[(4\hat{i} - 12\hat{j}) \times (2\hat{i} - 12\hat{j})] \\ &= -48\hat{k} \end{aligned}$$

8. (3) [Class 11th, Ch. 6, Page 96]



$$\therefore X_{CM} = \frac{m_1 x_1 + m_2 x_2}{m_1 + m_2}$$

$$\Rightarrow X_{CM} = \frac{20 \times 10}{20 + 10} = \frac{20}{3} \text{ m}$$

9. (3) [Class 11th, Ch. 3, Page 38]

$$u_x = (40)\cos(30^\circ)$$

$$= 40 \times \frac{\sqrt{3}}{2} = 20\sqrt{3} \text{ m/s}$$

At, $t = 2\text{ s}$ the body

is at maximum height

$$u_y = 0$$

10. (3) [Class 11th, Ch. 6, Page 114]

$$(A) I = \frac{ML^2}{12}$$

$$(B) I = \frac{(2M)L^2}{3} = \frac{2ML^2}{3}$$

$$(C) I = \frac{M(2L)^2}{12} = \frac{ML^2}{3}$$

$$(D) I = \frac{(2M)(2L)^2}{3} = \frac{8ML^2}{3}$$

11. (1) [Class 11th, Ch. 6, Page 114]

$$I = \frac{9MR^2}{2} - \left[\frac{M\left(\frac{R}{3}\right)^2}{2} + M\left(\frac{2R}{3}\right)^2 \right]$$

$$= 4MR^2$$

12. (3) [Class 11th, Ch. 5, Page 83]

After collision, speed will not be zero, $KE > 0$

13. (2) [Class 11th, Ch. 5, Page 74]

$$0 = (12 \times 4) - 4v \Rightarrow v = 12 \text{ m/s}$$

$$KE = \frac{1}{2}(4)(12)^2 = 288 \text{ J}$$

14. (3) [Class 11th, Ch. 4, Page 57]

Along x-axis momentum is conserved

$$mv + 0 = mv_x$$

$$v_x = v \quad \dots (1)$$

Along y-axis momentum conservation

$$0 + 0 = m\left(\frac{v}{\sqrt{3}}\right) - mv_y$$

$$v_y = \frac{v}{\sqrt{3}} \quad \dots (2)$$

$$v_{net} = \sqrt{v^2 + \frac{v^2}{3}} = \frac{2v}{\sqrt{3}}$$

15. (1) [Class 11th, Ch. 5, Page 75]

According to work and energy theorem

$$W_{Total} = k_f - k_i$$

$$p \times t = \frac{mv^2}{2p}$$

$$t = \frac{1}{2}mv^2 \dots (1)$$

$$v = C\sqrt{t}$$

$$S = c \int_0^t t^{1/2} dt$$

$$S \propto t^{3/2}$$

16. (1) [Class 11th, Ch. 4, Page 54]

Percentage change in momentum (P)

$$= \frac{P_2 - P_1}{P_1} \times 100$$

$$k = \frac{P^2}{2m}$$

$$P_2 = 2P_1 = 100\%$$

17. (2) [Class 11th, Ch. 7, Page 73]

$$W_{Porter} + W_{Gravity} = 0$$

$$W_{Porter} = -W_{Gravity}$$

$$= -mgh$$

$$= -80 \times 9.8 \times 0.8$$

$$= -627.2 \text{ J}$$

18. (1) [Class 11th, Ch. 4, Page 54]

$$\vec{F} = m\vec{a}$$

$$\vec{a} = \frac{d\vec{v}}{dt} = 2\hat{i} + 6\hat{j}$$

$$F = 0.5(2\hat{i} + 6\hat{j})$$

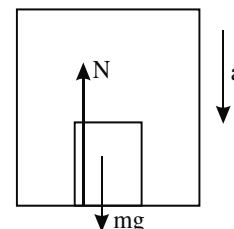
$$F = \hat{i} + 3\hat{j} \Rightarrow x = 3$$

19. (3) [Class 11th, Ch. 4, Page 56]

$$N = Mg / 4$$

$$Mg - N = Ma$$

$$a = \frac{3g}{4}$$



20. (2) [Class 11th, Ch. 6, Page 116]

$$\Rightarrow Mg \times \frac{L}{2} \sin 30^\circ = \frac{1}{2} \times I \times \omega^2$$

$$\Rightarrow Mg \frac{L}{2} \times \frac{1}{2} = \frac{1}{2} \times I \times \omega^2$$

$$\Rightarrow \frac{MgL \times 2}{4 \times I} = \omega^2$$

$$I = \frac{ML^2}{3}$$

$$\omega = \sqrt{30} \text{ rad/s}$$

21. (2) [Class 11th, Ch. 4, Page 56]

$$N = mg + 200 \times \sin 30^\circ$$

$$= 800 \text{ Newton}$$

22. (4) [Class 11th, Ch. 4, Page 60]

$$mg \sin \theta - \mu_k mg \cos \theta = ma$$

$$g \sin \theta - \mu_k \cos \theta = \frac{g}{4}$$

$$\mu_k = \frac{1}{2\sqrt{3}}$$

23. (3) [Class 11th, Ch. 7, Page 133]

For a sphere of mass M and radius R , the potential inside and outside of the sphere is

$$V_{\text{inside}} = -\frac{GM}{2R^3} [3R^2 - r^2]$$

r is distance from the center of sphere.

24. (4) [Class 11th, Ch. 6, Page 105]

$$\vec{\tau} = \vec{r} \times \vec{F}$$

$\vec{r}$ is perpendicular to $\vec{\tau}$

$\vec{F}$ is perpendicular to $\vec{\tau}$

25. (4) [Class 11th, Ch. 3, Page 38]

$$R = H$$

$$\frac{u^2 \sin 2\theta}{g} = \frac{2 \sin^2}{2}$$

26. (4) [Class 11th, Ch. 3, Page 38]

$$t_1 = \frac{2u \sin \theta}{g} \quad \frac{2 \cos}{g}$$

$$t_1 t_2 = \frac{2 \sin^2(2\theta)}{g^2} \quad \frac{2R}{g}$$

27. (4) [Class 11th, Ch. 3, Page 33]

$$\frac{dx}{dt} = Ky \quad \dots(1)$$

$$\frac{dy}{dt} = Kx \quad \dots(2)$$

$$\text{Divide (1) and (2)} \quad \frac{dx}{dy} = \frac{y}{x}$$

$$y = x^2 + \text{constant}$$

28. (2) [Class 11th, Ch. 6, Page 103]

$$\text{Average velocity} = \frac{R\sqrt{2}}{\frac{\pi R}{2v}} = \frac{2\sqrt{2}}{\pi}$$

$$\text{Instantaneous velocity} = v$$

29. (4) [Class 11th, Ch. 3, Page 34]

$$\dot{v}_1 = 5\hat{i}$$

$$\dot{v}_2 = 5\hat{j}$$

$$\Delta \dot{v} = 5\hat{j} - 5\hat{i} = 5(\hat{j} - \hat{i})$$

30. (3) [Class 11th, Ch. 2, Page 18]

Velocity and position is straight line so

$$\text{slope will be} = -\frac{v_0}{x_0}$$

So equation will be

$$v = mx + c$$

$$v = \frac{-v_0}{x_0} x \quad v_0 \quad \dots (i)$$

Acceleration will be

$$a = v \frac{dv}{dx}, \text{ so acceleration and velocity}$$

graph will be straight line.

31. (1) [Class 11th, Ch. 2, Page 18]

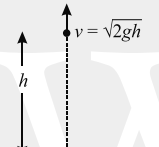
$$v^2 = u^2 + 2as \quad \dots(1)$$

$$v' = u^2 + 2a(s/2) \quad \dots(2)$$

$$= u^2 + \frac{2-u}{2}$$

$$v' = \sqrt{\frac{u^2 + 2}{2}}$$

32. (4) [Class 11th, Ch. 2, Page 19]



$$s = ut + \frac{1}{2}at^2 \quad \dots (i)$$

$$-h = \sqrt{2gh}t - \frac{1}{2}gt^2$$

$$\frac{1}{2}gt^2 - \sqrt{2gh}t - h = 0$$

$$t = \frac{\sqrt{2gh} \pm \sqrt{2gh + 4h\left(\frac{g}{2}\right)}}{2\left(\frac{g}{2}\right)}$$

$$t = \frac{\sqrt{2gh} \pm \sqrt{2gh + 2gh}}{g}$$

$$t = \frac{\sqrt{2gh}}{g} [1 + \sqrt{2}]$$

33. (2) [Class 11th, Ch. 3, Page 34]

$$v_3 = 150 - 30 = 120$$

$$v_5 = 150 - 50 = 100$$

$$\frac{x+1}{x} = \frac{5}{5}$$

34. (4) [Class 11th, Ch. 3, Page 37]

Suppose total time taken is t and height h then

$$h = \frac{1}{2}gt^2 \quad \dots(1)$$

Covering first half distance, time taken

$$\frac{h}{2} = \frac{1}{2}gt_1^2 \quad \dots(2)$$

$$t_1 = \sqrt{\frac{h}{g}}$$

$$t_2 = t - t_1$$

$$= t - \sqrt{\frac{h}{g}}$$

$$t_2 = \sqrt{\frac{2h}{g}} \quad \sqrt{-} = \sqrt{\frac{h}{g}} (\sqrt{2} - 1)$$

$$t_2 = t (\sqrt{2} - 1)$$

35. (1)

$$\bar{T} = \frac{T_1 + T_2 + T_3 + T_4}{4} \quad 92 \text{ sec}$$

$$\bar{T} = \frac{2 + 1 + 3 + 0}{4} \quad 1.5$$

36. (3) [Class 11th, Ch. 1, Page 7]

$$\frac{1}{2}\epsilon_0 E^2 \equiv \text{electrical energy density}$$

$$= \frac{\text{Energy}}{\text{Volume}}$$

37. (2) [Class 11th, Ch. 1, Page 6]

$$\frac{dX}{X} = 2 \frac{dA}{A} + \frac{1}{2} \frac{dB}{B} - \frac{dC}{C} + \frac{dD}{D}$$

38. (2) [Class 11th, Ch. 2, Page 18]

$$u = 0$$

$$h = \frac{1}{2}gt^2$$

Distance covered while going upwards in time t = distance covered while coming downward in time t .

39. (1) [Class 11th, Ch. 7, Page 132]

Outside of the earth

$$g \propto \frac{1}{r^2}$$

Inside the earth

$$g \propto r$$

40. (3) [Class 11th, Ch. 3, Page 30]

$$dx_{cm} = \frac{m_1 dx_1 + m_2 dx_2}{m_1 + m_2} \quad \dots(1)$$

$$dx_{cm} = 0$$

$$m_1 dx_1 + m_2 dx_2 = 0 \quad \dots(2)$$

$$60 = -30dx_2$$

$$dx_2 = -2 \text{ cm}$$

41. (3) [Class 11th, Ch. 1, Page 7]

$$[\text{Pressure}][\text{Time}] = \left[\frac{MLT^{-2}}{L^2} \right] [T]$$

$$[ML^{-1}T^{-1}] \dots (i)$$

$$F = 6\pi\eta rv$$

$$[\eta] = \left[\frac{F}{rv} \right] = \left[\frac{MLT^{-2}}{L^2T^{-1}} \right]$$

$$= [ML^{-1}T^{-1}]$$

42. (2) [Class 11th, Ch. 7, Page 128]

According to kepler's law, in equal time, equal area will be covered. Since area is doubled in SCD time will be double.

$$\text{Area (SCD)} = 2 \text{ Area (SAB)}$$

$$t_1 = 2t_2$$

43. (1) [Class 11th, Ch. 3, Page 48]

Given that, radius,

$$R = 0.1 \text{ m}$$

Angular speed,

$$\omega = \frac{2\pi}{T} = \frac{2\pi}{60} = 0.105 \text{ rad/s}$$

$\Rightarrow$ Average acceleration,

$$a = \omega^2 R = (0.105)^2 (0.1)$$

$$= 1.102 \times 10^{-3} \text{ m/s}^2$$

and it is of the order of 10^{-3} .

44. (1) [Class 11th, Ch. 4, Page 60]

$$F \cos 30^\circ = \mu(mg - F \sin 30^\circ)$$

$$F \cos 30^\circ = \mu mg - \mu F \sin 30^\circ$$

$$F(\cos 30^\circ + \mu \sin 30^\circ) = \mu mg$$

$$F = \frac{\mu mg}{\cos 30^\circ + \mu \sin 30^\circ}$$

45. (1) [Class 11th, Ch. 5, Page 54]

Momentum would be maximum, when KE would be maximum and this is the case when total elastic PE is converted into KE.

According to conservation of energy,

$$\frac{1}{2} kL^2 = \frac{1}{2} Mv^2$$

$$\Rightarrow kL^2 = \frac{(Mv)^2}{M}$$

$$\text{or } MkL^2 = p^2$$

$$\Rightarrow p = L\sqrt{Mk}$$

Chemistry

46. (4) [Class 11th, Ch. 2, Page 41]

For each metal, there is a characteristic minimum frequency, ν_0 (also known as threshold frequency) below which photoelectric effect is not observed. At a frequency $\nu > \nu_0$, the ejected electrons come out with certain kinetic energy. Threshold frequency depends on the work function of metal.

47. (3) [Class 11th, Ch. 1, Page 18]

$$\text{Number of moles of } H_2O = \frac{36}{18} = 2$$

moles

H has zero neutron and O has 8 neutrons.

Number of neutrons in 2 moles of

$$H_2O = 2 \times 8 \times N_A = 16 N_A$$

48. (2) [Class 11th, Ch. 3, Page 90]

Order of negative electron gain enthalpy is:

$$S > Se > Te > O$$

49. (2) [Class 11th, Ch. 4, Page 113]

The repulsive interaction of electron pair decreases in the order:

Lone pair – lone pair > lone pair – bond pair > bond pair – bond pair.

50. (4) [Class 11th, Ch. 6, Page 203]

For basic buffer,

$$pOH = pK_b + \log \frac{[\text{salt}]}{[\text{base}]}$$

$$pOH = -\log K_b + \log \frac{[\text{salt}]}{[\text{base}]}$$

$$pOH = -\log(1 \times 10^{-5}) + \log 1 = 5$$

$$pH + pOH = 14$$

$$pH = 14 - 5 = 9$$

51. (2) [Class 11th, Ch. 5, Page 141]

$$w = -P_{\text{ext}} \times \Delta V$$

$$w = -1 \times 0.8 = -0.8 \text{ L atm}$$

$$\text{Since, } 1 \text{ L atm} = 101.325 \text{ J}$$

$$\text{So, } w = -0.8 \times 101.325 = -81.06 \text{ J}$$

52. (2) [Class 11th, Ch. 2, Page 45]

In Balmer series, transition of electron takes place from higher orbits to $n_1 = 2$.

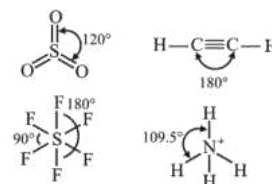
If transition of electron takes place from 6th to 3rd orbit, then the total

number of spectral lines in Balmer series is 0.

53. (3) [Class 11th, Ch. 3, Page 85]

Non-metallic character increases as we move left to right across the period. Hence, the element with electronic configuration $1s^2 2s^2 2p^6 3s^2 3p^5$ will have maximum non-metallic character.

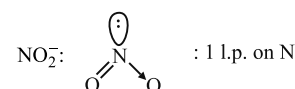
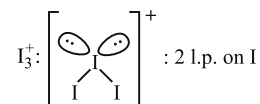
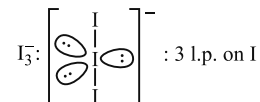
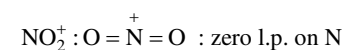
54. (1) [Class 11th, Ch. 4, Page 115]



55. (1) [Class 11th, Ch. 1, Page 12]

Zeros preceding to first non-zero digit are not significant. Thus, number of significant figures present in 151 cm, 0.151 cm and 0.0151 cm are 3, 3 and 3 respectively.

56. (1) [Class 11th, Ch. 4, Page 116]



57. (4) [Class 11th, Ch. 1, Page 16]

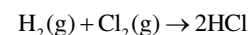
(B) Matter consists of indivisible atoms.

(D) All the atoms of given element have same properties including mass.

58. (3) [Class 11th, Ch. 1, Page 20]

$$\text{Moles of } H_2 = \frac{2.24}{22.4} = 0.1$$

$$\text{Moles of } Cl_2 = \frac{0.112}{22.4} = 0.005$$



Initial moles: 0.1 0.005

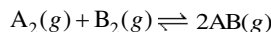
Here, Cl_2 is limiting reagent.

1 mole of Cl_2 produces 2 moles of HCl.

0.005 moles of Cl_2 produces 0.01 moles of HCl .

59. (1) [Class 11th, Ch. 2, Page 32]

Anode rays or canal rays are made up of positively charged particles and the characteristics of anode rays depends upon the material of electrodes and the nature of gas present in the cathode ray tube.

60. (1) [Class 11th, Ch. 6, Page 178]

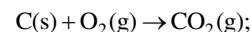
$$\text{Initial conc.} \quad \frac{5}{2} \quad \frac{4}{2} \quad \frac{6}{2}$$

$$\text{at eq.} \quad 2.5 - x \quad 2 - x \quad 3 + 2x$$

$$\therefore 3 + 2x = 5$$

$$\Rightarrow x = 1$$

$$K_c = \frac{(5)^2}{1.5 \times 1} = 16.66$$

61. (4) [Class 11th, Ch. 5, Page 152]

$\therefore$ Heat released on formation of 44 g CO_2

$$= -3.935 \times 10^5 \text{ J/mol}$$

$\therefore$ Heat released on formation of 88 g CO_2

$$= \frac{-3.935 \times 10^5}{44} \times 88 = -7.87 \times 10^5 \text{ J/mol}$$

$$= -787 \text{ kJ/mol}$$

62. (1) [Class 11th, Ch. 5, Page 138]

Path function are those which depends on the initial and final state of the system.

63. (1) [Class 11th, Ch. 6, Page 169]

Volatile liquids have higher vapour pressure than other liquid at the same temperature. More volatile liquid has lower boiling point and hence can be vaporised easily.

64. (4) [Class 11th, Ch. 1, Page 19]

Gay Lussac's law deals only with gaseous reactants and products at same temperature and pressure.

65. (3) [Class 11th, Ch. 6, Page 178]

$$\frac{K_p}{K_c} = (\text{RT})^{\Delta n_g}$$

$\frac{K_p}{K_c}$ ratio will be minimum when Δn_g is minimum.

66. (1) [Class 11th, Ch. 2, Page 59]

No. of angular nodes = l

For 3s-orbitals, $l = 0$

For 4p-orbitals, $l = 1$

For 3d-orbitals, $l = 2$

For 5f-orbitals, $l = 3$

67. (3) [Class 11th, Ch. 5, Page 143]

In cyclic process ΔU and ΔH is zero and when initial and final state of a system in a process are the same the process is called a cyclic process.

68. (1) [Class 11th, Ch. 4, Page 129]

Bond order of $\text{N}_2 = 3$

Bond order of $\text{O}_2^{2+} = 3$

Bond order of $\text{O}_2^- = 1.5$

Bond order of $\text{O}_2^{2-} = 1$

Bond order of $\text{O}_2^+ = 2.5$

69. (4) [Class 11th, Ch. 1, Page 20]

For the reaction,



$$\text{Moles of } \text{CO}_2 = \frac{1.12}{22.4} = 0.05 \text{ mol}$$

1 mol of CO_2 is produced by 1 mole of CaCO_3 .

0.05 mole of CO_2 will be produced by 0.05 mole of CaCO_3 .

$$\text{Mass of pure } \text{CaCO}_3 = 0.05 \times 100 = 5 \text{ g}$$

Hence, amount of CaCO_3 required

$$= \frac{5}{25} \times 100 = 20 \text{ g}$$

70. (1) [Class 11th, Ch. 3, Page 87]

In isoelectronic species, more the negative charge on the ion, more will be the size and more the positive charge on the ion, lesser will be the size.

Hence, the correct order of size is :

**71. (2) [Class 11th, Ch. 2, Page 42]**

$E = W + K.E$ (W is the work function)

$$8.4 = 1.4 + K.E$$

$$K.E = 7 \text{ eV}$$

72. (2) [Class 11th, Ch. 4, Page 129]

Energy order for the molecular orbital diagram of F_2 :

$$\sigma 1s < \sigma^* 1s < \sigma 2s < \sigma^* 2s < \sigma 2p_z < (\pi 2p_x = \pi 2p_y) < (\pi^* 2p_x = \pi^* 2p_y) < \sigma^* 2p_z$$

73. (1) [Class 11th, Ch. 2, Page 55]

$$n = 4, l = +2$$

Possible value of $m = -l$ to $+l = -2, -1, 0, +1, +2$

But $m = 0$ (given)

Only 1 orbital which has $m = 0$

74. (2) [Class 11th, Ch. 5, Page 145]

$$Q = nC\Delta T = \frac{108}{27} \times 24 \times 20 \text{ J} = 1.92 \text{ kJ}$$

75. (2) [Class 11th, Ch. 2, Page 35]

For a neutral atom, number of electrons are equal to the number of protons.

For the given ion, number of electrons is 2 more than the number of protons. So, the charge on the ion is -2 .

76. (3) [Class 11th, Ch. 4, Page 112]

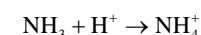
- Smaller the size of the cation and larger the size of anion, greater is the covalent character of an ionic bond.
- Greater the charge of cation, greater the covalent character of the ionic bond.

77. (4) [Class 11th, Ch. 4, Page 114]

SF_6 molecule is octahedral and dipole moment is zero.

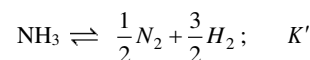
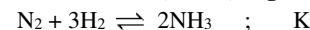
78. (2) [Class 11th, Ch. 1, Page 24]

Temperature is directly proportional to Volume, so if temperature changes, it will affect the volume and thus molarity. Mole fraction, Molality and Weight percent are not temperature dependent since, they do not contain any volume term in them.

79. (2) [Class 11th, Ch. 6, Page 190]

(base) (conjugate acid)

K_w increases with increase in temperature.

80. (3) [Class 11th, Ch. 6, Page 181]

New equilibrium constant K' will become:

$$K' = \frac{1}{\sqrt{K}} = \frac{1}{\sqrt{25}} = \frac{1}{5} = 0.2$$

81. (4) [Class 11th, Ch. 3, Page 82]

Nitrogen belongs to 2nd period and 15th group.

Beryllium belongs to 2nd period and 2nd group.
Scandium belongs to 4th period and 3rd group.
Chlorine belongs to 3rd period and 17th group.

82. (2) [Class 11th, Ch. 5, Page 144]

Electrode potential does not depend on the size (mass) of a system so it is an intensive property.

83. (3) [Class 11th, Ch. 2, Page 55]

- For any given value of n , the value of ' l ' lies between 0 to $(n-1)$, but $n \neq l$.
- The value of m lies between $-l$ to $+l$. For $l = 1$, m may be $-1, 0, +1$ only.
- Hence, $n = 3, l = 1, m = -2, s = -\frac{1}{2}$ is

not possible for an electron.

84. (3) [Class 11th, Ch. 6, Page 178]

The gaseous mixture contains 40% Cl_2 and 40% PCl_3 , since they are produced in 1:1 mole ratio.

PCl_5 is 20.

For ideal gases, mole% = volume%

$$P_{\text{Cl}_2} = P_{\text{PCl}_3}$$

$$2 \times 0.40 = 0.8 \text{ atm}$$

$$P_{\text{PCl}_5} = 2 \times 0.2 = 0.40 \text{ atm}$$

$$K_p = \frac{P_{\text{PCl}_3} \times P_{\text{Cl}_2}}{P_{\text{PCl}_5}} = \frac{0.80 \times 0.80}{0.40} = 1.6 \text{ atm}$$

85. (1) [Class 11th, Ch. 1, Page 20]

Molecular weight = $2 \times$ Vapour density
 $= 2 \times 11.2 = 22.4$

$\therefore 22.4 \text{ g of gas occupies } 22.4 \text{ L at STP}$

$$\therefore 5.2 \text{ g of gas occupies} = \frac{22.4}{22.4} \times 5.2 = 5.2 \text{ L}$$

86. (3) [Class 11th, Ch. 3, Page 78]

Henry Moseley plotted $\sqrt{\nu}$ (square root of the frequency of X-rays emitted) against atomic number (Z), obtained a straight line.

87. (4) [Class 11th, Ch. 3, Page 84]

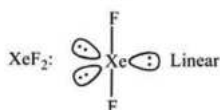
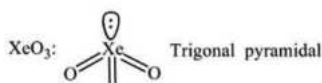
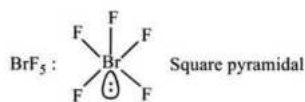
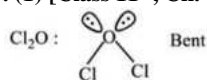
Block	General outer electronic configuration
-------	--

s	ns^{1-2}
p	$ns^2 np^{1-6}$
d	$(n-1)d^{1-10} ns^{1-2}$
f	$(n-2)f^{1-14} (n-1)d^{0-1} ns^2$

88. (2) [Class 11th, Ch. 3, Page 86]

Atomic radius of the elements increases as one moves down the first group of the periodic table.

89. (1) [Class 11th, Ch. 4, Page 115]



90. (2) [Class 11th, Ch. 2, Page 48]

$$\text{Radius of } n\text{th orbit, } r_n = 0.529 \frac{n^2}{Z} \text{ \AA}$$

n = number of orbit, Z = atomic number
For 1st orbit of Be^{3+} , $n = 1, Z = 4$

$$r_1 = 0.529 \frac{1^2}{4} = 0.132 \text{ \AA}$$

Botany

91. (4) [Class 11th, Ch. 1, Page 04]

Generic name starts with a capital letter.

92. (4) [Class 11th, Ch. 2, Page 21]

The absence of a protein coat (stated in the Reason) is a characteristic of viroids, but it is not the direct reason why viroids are free RNA molecules.

93. (1) [Class 11th, Ch. 3, Page 24]

Fusion between one large, nonmotile (static) female gamete and a smaller, motile male gamete is termed oogamous, e.g., *Volvox*, *Fucus*.

94. (2) [Class 11th, Ch. 3, Page 24]

Numerical taxonomy which is now easily carried out using computers is based on all observable characteristics. Number and codes are assigned to all the characters and the data are then processed. In this way each character is given equal importance and at the same time hundreds of characters can be considered. Cytotaxonomy that is based on cytological information like chromosome number, structure, behaviour and chemotaxonomy that uses the chemical constituents of the plant to resolve confusions, are also used by taxonomists these days.

95. (4) [Class 11th, Ch. 2, Page 16, 17]

- Asexual reproduction in fungi is by spores called conidia or sporangiospores or zoospores.
- Fusion of protoplasm between two motile or non-motile gametes called plasmogamy and fusion of two nuclei is called karyogamy.

96. (1) [Class 11th, Ch. 5, Page 62]

A flower is asymmetric (irregular) if it cannot be divided into two similar halves by any vertical plane passing through the centre, as in canna.

97. (2) [Class 11th, Ch. 5, Page 67]

The outer covering of endosperm separates the embryo by a proteinous layer called aleurone layer.

98. (2) [Class 11th, Ch. 1, Page 08]

Family	Muscidae
Phylum	Arthropoda
Order	Diptera
Species	<i>Domestica</i>
Genus	<i>Musca</i>

99. (4) [Class 11th, Ch. 3, Page 27]

The gametes of brown algae (phaeophyceae) are pyriform (pear-shaped) and bear two laterally attached flagella.

100. (3) [Class 11th, Ch. 3, Page 32]

In pteridophytes, the female gametophytes are retained on the parent sporophytes for variable periods. The development of the zygotes into young embryos take place within the female gametophytes. This event is a precursor to the seed habit considered an important step in evolution.

101. (3) [Class 11th, Ch. 3, Page 27]

- The red thalli of most of the red algae are multicellular. Some of them have complex body organisation. The food is stored as floridean starch which is very similar to amylopectin and glycogen in structure.
- Majority of the red algae are marine with greater concentrations found in the warmer areas. They occur in both well-lighted regions close to the surface of water and also at great depths in oceans where relatively little light penetrates.

102. (3) [Class 11th, Ch. 3, Page 30, 33]

- Unlike bryophytes and pteridophytes, in gymnosperms the male and the female gametophytes do not have an independent free-living existence.

103. (4) [Class 11th, Ch. 3, Page 27]

The major pigments in the class of algae that store their food in the form of floridean starch i.e., red algae are chlorophyll a, d and phycoerythrin.

104. (3) [Class 11th, Ch. 5, Page 65]

In parietal placentation, the ovules develop on the inner wall of the ovary or on peripheral part. Ovary is one-chambered but it becomes two-chambered due to the formation of the false septum, e.g., mustard and *Argemone*.

105. (2) [Class 11th, Ch. 5, Page 67, 68]

Tap roots of carrot, turnip and adventitious roots of sweet potato, get swollen and store food.

106. (3) [Class 11th, Ch. 5, Page 61]

In racemose type of inflorescences the main axis continues to grow, the flowers are borne laterally in an acropetal succession.

107. (2) [Class 11th, Ch. 1, Page 04, 05, 08]

- Brain-dead human patients do not have self-consciousness.
- Each rank or taxon represents a unit of classification.

108. (1) [Class 11th, Ch. 2, Page 13]

Archaeobacteria differ from other bacteria in having a different cell wall structure.

109. (2) [Class 11th, Ch. 3, Page 32]

Roots in some genera have fungal association in the form of mycorrhiza (*Pinus*), while in some others (*Cycas*) small specialised roots called coralloid roots are associated with N₂-fixing cyanobacteria. The stems are unbranched (*Cycas*) or branched (*Pinus*, *Cedrus*).

110. (1) [Class 11th, Ch. 2, Page 14, 15]

Plasmodium	Slime moulds
Mixotrophic nutrition	Euglenoids
Red tide	Dinoflagellates
Oceanic chief producers	Diatoms

111. (4) [Class 11th, Ch. 2, Page 20]

In addition to proteins, viruses also contain genetic material, that could be either RNA or DNA. No virus contains both RNA and DNA.

112. (3) [Class 11th, Ch. 5, Page 62, 63]

If gynoecium is situated in the centre and other parts of the flower are located on the rim of the thalamus almost at the same level, it is called perigynous. The ovary here is said to be half inferior, e.g., plum, rose, peach.

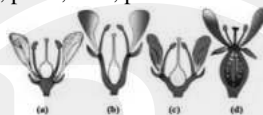


Figure: Position of floral parts on thalamus:

(a) Hypogynous (b) and (c) Perigynous (d) Epigynous

113. (4) [Class 11th, Ch. 6, Page 74]

The innermost layer of the cortex is called endodermis. It comprises a single layer of barrel shaped cells without any intercellular spaces. The tangential as well as radial walls of the endodermal cells have a deposition of water-impermeable, waxy material suberin in the form of casparian strips.

114. (1) [Class 11th, Ch. 5, Page 66]

In mango, the pericarp is well differentiated into an outer thin epicarp, a middle fleshy edible mesocarp and an inner stony hard endocarp.

115. (4) [Class 11th, Ch. 6, Page 77]

In an isobilateral leaf, the stomata are present on both the surfaces of the epidermis; and the mesophyll is not differentiated into palisade and spongy parenchyma.

116. (3) [Class 11th, Ch. 5, Page 63]

Cassia and gulmohur both exhibit imbricate aestivation, where the margins of sepals or petals overlap one another but not in any particular direction.

117. (1) [Class 11th, Ch. 6, Page 74]

The internal tissue organisation in dicot root are as follows:

The outermost layer is epiblema. Many of the cells of epiblema protrude in the form of unicellular root hairs. The cortex consists of several layers of thin-walled parenchyma cells with intercellular spaces. The innermost layer of the cortex is called endodermis.

118. (2) [Class 11th, Ch. 5, Page 67, 68]

The floral diagram and floral formula in Figure represents the mustard plant (Family: Cruciferae or Brassicaceae).

**119. (2) [Class 11th, Ch. 6, Page 74]**

All tissues on the innerside of the endodermis such as pericycle, vascular bundles and pith constitute the stele.

120. (3) [Class 11th, Ch. 2, Page 18]

Agaricus (mushroom) is a member of basidiomycetes.

121. (1) [Class 11th, Ch. 6, Page 64]

In a typical dicot stem, from the inside to outside, the sequence of layers is: Phloem, pericycle, endodermis, parenchyma, collenchyma, epidermis.

122. (1) [Class 11th, Ch. 6, Page 76]

Mesophyll, which possesses chloroplasts and carry out photosynthesis, is made up of parenchyma

123. (1) [Class 11th, Ch. 5, Page 64, 65]

- If gynoecium is situated in the centre and other parts of the flower are located on the rim of the thalamus almost at the same level, it is called perigynous. The ovary here is said to be half inferior, e.g., plum, rose, peach.
- When a flower can be divided into two similar halves only in one particular vertical plane, it is zygomorphic, e.g., pea, gulmohur, bean, *Cassia*.
- The term syncarpous is a condition when carpels are fused, as in mustard and tomato.

- When stamens are attached to the petals, they are epipetalous as in brinjal, or epiphyllous when attached to the perianth as in the flowers of lily.

124. (2) [Class 11th, Ch. 2, Page 17, 18]

<i>Puccinia</i>	wheat rust pathogen
<i>Neurospora</i>	used in biochemical & genetic work
Saprophytes	feed on dead substrates
<i>Albugo</i>	parasitic fungus on mustard

125. (3) [Class 11th, Ch. 5, Page 64]

(Type of aestivation)	(Example)
Valvate – Sepals or petals in a whorl just touch one another at the margin, without overlapping.	<i>Calotropis</i>
Imbricate – The margins of sepals or petals overlap one another but not in any particular direction.	<i>Cassia</i>
Vexillary – Five petals, the largest (standard) overlaps the two lateral petals (wings) which in turn overlap the two smallest anterior petals (keel).	Bean

126. (4) [Class 11th, Ch. 5, Page 62, 66]

- When a flower can be divided into two equal radial halves in any radial plane passing through the centre, it is said to be actinomorphic, e.g., mustard, *datura*, chili.

- Maize is a monocot plant.

127. (2) [Class 11th, Ch. 6, Page 74]

The parenchymatous cells which lie between the xylem and the phloem are called conjunctive tissue.

128. (4) [Class 11th, Ch. 6, Page 76]

In dicot stem, a large number of vascular bundles are arranged in a ring; the 'ring' arrangement of vascular bundles is a characteristic of dicot stem. Each vascular bundle is conjoint, open, and with endarch protoxylem.

129. (3) [Class 11th, Ch. 5, Page 59, 60]

- A typical leaf consists of three main parts: leaf base, petiole and lamina. The leaf is attached to the stem by the leaf base. In some leguminous plants the leaf base may become swollen, which is called the pulvinus. The lamina or the leaf blade is the green expanded part of the leaf with veins and veinlets.
- The root is covered at the apex by a thimble-like structure called the root cap.

130. (4) [Class 11th, Ch. 6, Page 72]

The outside of the epidermis is often covered with a waxy thick layer called the cuticle which prevents the loss of water. Cuticle is absent in roots. Stomata are structures present in the epidermis of leaves. Stomata regulate the process of transpiration and gaseous exchange. Each stoma is composed of two bean shaped cells known as guard cells which enclose stomatal pore. The guard cells possess chloroplasts and regulate the opening and closing of stomata. Sometimes, a few epidermal cells, in the vicinity of the guard cells become specialised in their shape and size and are known as subsidiary cells.

131. (2)

- Statement A: Incorrect. *Triticum aestivum*, commonly known as wheat, has a fibrous root system rather than an adventitious root system with stilt roots.
- Statement D: Incorrect. Wheat flowers are perfect (bisexual) rather than separate male and female flowers; they typically have a bicarpellary, syncarpous gynoecium.

132. (2) [Class 11th, Ch. 5, Page 65]

	Axile
	Parietal
	Free central
	Basal

133. (4) [Class 11th, Ch. 6, Page 72]

The trichomes in the shoot system are usually multicellular. They may be branched or unbranched and soft or stiff. They may even be secretory. The trichomes help in preventing water loss due to transpiration.

134. (1) [Class 11th, Ch. 5, Page 67]

Aleurone layer – The outer covering of endosperm separates the embryo by a proteinous layer called aleurone layer. The embryo is small and situated in a groove at one end of the endosperm. It consists of one large and shield shaped cotyledon known as scutellum and a short axis with a plumule and a radicle. The plumule and radicle are enclosed in sheaths which are called coleoptile and coleorhiza respectively.

135. (3) [Class 11th, Ch. 6, Page 72]

The inner wall of each guard cell, towards the pore or stomatal aperture, is thick.

Zoology

136. (4) [Class 11th, Ch. 4, Page 45]

The excretory organ of hemichordates is the proboscis gland.

137. (1)

- In male cockroaches, a mushroom-shaped gland is present in the 6th-7th abdominal segments, which functions as an accessory reproductive gland.
- Cockroaches are uricotelic.
- The urecose glands help in excretion.
- Blood from the sinuses enters the heart through the ostia and is pumped anteriorly to the sinuses again.

138. (1) [Class 11th, Ch. 9, Page 108, 109]

Macromolecules in the acid-insoluble fraction include proteins, nucleic acids, and polysaccharides. Lipids, although present, are not strictly considered macromolecules in this context.

139. (3) [Class 11th, Ch. 4, Page 47, 48]

<i>Myxine</i>	Hagfish
<i>Pristis</i>	Sawfish
<i>Exocoetus</i>	Flying fish
<i>Betta</i>	Fighting fish

140. (3)

- Exocrine glands secrete mucus, saliva, earwax, oil, milk, digestive enzymes and other cell products. These products are released through ducts or tubes.
- In contrast, endocrine glands do not have ducts. Their products called

hormones are secreted directly into the fluid bathing the gland.

- Insulin is a hormone that decreases the blood glucose levels.

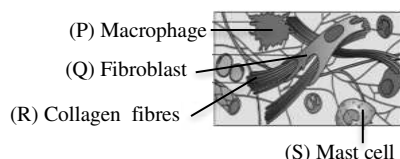
141. (2) [Class 11th, Ch. 9, Page 109]

Lipids, though have a small molecular weight as compared to biomacromolecules come under acid insoluble fraction because they form vesicles that are insoluble in water.

142. (1) [Class 11th, Ch. 4, Page 45, 47]

- Cyclostomes have a sucking and circular mouth without jaws.
- Hemichordates have a rudimentary structure in the collar region called stomochord.

143. (3)



144. (4) [Class 11th, Ch. 9, Page 109, 111]

- GLUT-4 enables glucose transport into the cells.
- Water is the most abundant chemical in the living organism.
- Paper and cotton fibre are cellulosic in nature.
- Chitin is a complex polysaccharide.

145. (2) [Class 11th, Ch. 4, Page 45]

Chaetopleura (Chiton) belongs to the phylum Mollusca.

146. (1)

Loose connective tissue	Adipose tissue
Dense regular connective tissue	Tendons
Dense irregular connective tissue	Skin
Specialised connective tissue	Cartilage

147. (1) [Class 11th, Ch. 9, Page 117]

Malonate acts as competitive inhibitor for succinic dehydrogenase because it closely resembles the substrate succinate in structure.

148. (1) [Class 11th, Ch. 4, Page 38]

- Cnidarians are diploblastic.

- In diploblastic organisms, cells are arranged in two embryonic layers, an external ectoderm and an inner endoderm.

149. (3)

- The smooth muscle fibres taper at both ends (fusiform) and do not show striations.
- Smooth muscles are 'involuntary' as their functioning cannot be directly controlled.

150. (3) [Class 11th, Ch. 4, Page 45, 47]

- Ctenophores are exclusively marine and diploblastic organisms.
- Bioluminescence is well-marked in ctenophores.

151. (2) [Class 11th, Ch. 9, Page 106]

- Uracil + Sugar molecules = Uridine
- Uridine + Phosphate group = Uridylic acid

152. (4)

- Connective tissues are specialised for linking and connecting other tissues/organs.
- Ligament is a the type of dense regular connective tissue.

153. (2) [Class 11th, Ch. 9, Page 106]

- Lecithin is a type of phospholipid, which is an essential component of cell membranes.

154. (3) [Class 11th, Ch. 4, Page 48]

- Amphibians have external fertilisation and indirect development.
- The skin of amphibians is moist and without scales.
- The body of an amphibian is divided into a head and a trunk.
- Amphibians have three-chambered hearts.

155. (4)

Chondrocytes are cells of cartilage.

156. (2) [Class 11th, Ch. 9, Page 109]

The protein portion of the enzymes	Apoenzyme
Possess peptide bonds	Collagen
Cell wall material in plants	Cellulose
Secondary metabolite	Ricin

157. (4) [Class 11th, Ch. 4, Page 45]

- *Asterias* (Starfish) belong to the phylum Echinodermata.
- The adult echinoderms are radially symmetrical but their larvae are bilaterally symmetrical.

158. (1)

- Cell junctions are present in the epithelial tissue.
- Different types of cell junctions are present in animal tissue.

159. (2) [Class 11th, Ch. 9, Page 106]

Palmitic acid has 16 carbons including carboxyl carbon.

160. (3) [Class 11th, Ch. 4, Page 42]

Spongocoel	Porifera
Dorso-ventrally flattened body	Platyhelminthes
Coelenteron cavity	Cnidaria
Chitinous exoskeleton	Arthropoda

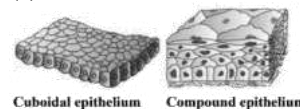
161. (1)

Communication junctions (intercalated discs) at some fusion points allow the cells to contract as a unit.

162. (1) [Class 11th, Ch. 9, Page 112]

The catalytic cycle begins with the substrate binding to the enzyme's active site (b). This binding induces the enzyme to alter its shape for a tighter fit (c). Subsequently, the active site breaks the substrate's bonds, creating an enzyme-product complex (d). Finally, the enzyme releases the products and is ready to start the process over with a new substrate molecule (a).

163. (3)



- The **cuboidal epithelium (A)** is composed of a single layer of cube-like cells. This is commonly found in ducts of glands and tubular parts of nephrons in kidneys and its main functions are secretion and absorption. The epithelium of proximal convoluted tubule (PCT) of nephron in the kidney has microvilli.
- **Compound epithelium (B)** is made of more than one layer (multi-layered) of cells and thus has a limited role in secretion and absorption. Their main function is to provide protection

against chemical and mechanical stresses. They cover the dry surface of the skin, the moist surface of buccal cavity, pharynx, inner lining of ducts of salivary glands and of pancreatic ducts.

164. (2) [Class 11th, Ch. 4, Page 43, 44]

- Starfish belongs to the phylum Echinodermata and has a water vascular system that helps in the locomotion, capture and transport of food and respiration.
- King crab (*Limulus*) has book gills, that help in respiration.
- Cuttlefish belong to the phylum Mollusca; in them features like gills are present in the mantle cavity that helps in respiration and excretion.
- Earthworm respire through the moist skin.

165. (4) [Class 11th, Ch. 9, Page 110]

In glycogen, the right end is reducing and the left end is non-reducing.

166. (2)

In cockroaches, the sense organs are antennae, eyes, maxillary palps, labial palps, anal cerci, etc.

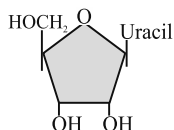
Phallomere and anal styles are not sensory organs; phallomere is part of the male reproductive structure, and anal styles aid in copulation and are not sensory.

167. (2) [Class 11th, Ch. 4, Page 43, 44]

Radula	Rasping organ
Parapodia	Helps in swimming
Flame cells	Helps in excretion
Gills	Respiratory organ

168. (4) [Class 11th, Ch. 9, Page 107]

- When nitrogenous bases like adenine, guanine, cytosine, thymine and uracil are attached to a sugar, they are called nucleosides.



- The given figure shows uridine (uracil + sugar) is a nucleoside.

169. (3)

- Bones have a hard and non-pliable ground substance.
- The matrix of bone is rich in calcium salt and collagen fibres (give strength to bones).
- Bone marrow in some bones is the site of the production of blood cells.
- The bone cells are called osteocytes.

170. (3) [Class 11th, Ch. 4, Page 44]

Vector	<i>Anopheles</i>
Economically important insect	<i>Bombyx</i>
Gregarious pest	<i>Locusta</i>
Living fossil	<i>Limulus</i>

171. (3) [Class 11th, Ch. 9, Page 110]

Cellulose is a homopolymer because it is made of several units of glucose.

172. (2)

- Neuroglia make up more than one-half of the volume of neural tissue.
- Tight junctions help to stop the leakage of substances across a tissue.

173. (3) [Class 11th, Ch. 4, Page 46]



The given diagram shows an organism, *Ascidia*. It belongs to the subphylum Urochordata and they are exclusively marine.

174. (2) [Class 11th, Ch. 9, Page 104]

Trichloroacetic acid (Cl_3CCOOH) is used for chemical analysis of living tissues.

175. (1) [Class 11th, Ch. 7, Page 80]

Camouflage	Ability to change colour to hide from their enemies
Aestivation	Summer sleep
Hibernation	Winter sleep
Poikilotherm	Cold-blooded

176. (4) [Class 11th, Ch. 4, Page 41]

- Polyps produce medusa asexually.
- Metamerism is seen in earthworms.
- Medusa forms polyps sexually.
- Corals have a skeleton composed of calcium carbonate.

177. (4) [Class 11th, Ch. 9, Page 113, 114]

- Inorganic compounds like sulphate, phosphate, etc., are seen in the acid-soluble fraction.
- Arachidonic acid has 20 carbon atoms including carboxyl carbon.

178. (2) [Class 11th, Ch. 4, Page 97]

Cyclostome migrate to freshwater(b) for spawning (a). Soon after spawning, the parent cyclostome dies (c). The larvae undergo metamorphosis (e) and eventually return to the ocean (d).

179. (4) [Class 11th, Ch. 9, Page 146]

- Acid insoluble fraction has only four types of organic compounds.
- Biomacromolecules have a molecular weight in the range of 10,000 Da and above.
- Molecular weights less than one thousand Dalton are usually referred to as micromolecules.
- All the carbon-containing compounds in a living tissue are called biomolecules.

180. (3) [Class 11th, Ch. 4, Page 46]

- An open circulatory system is found in most arthropods.
- In Aves, skin is dry without glands except the oil gland at the base of the tail.

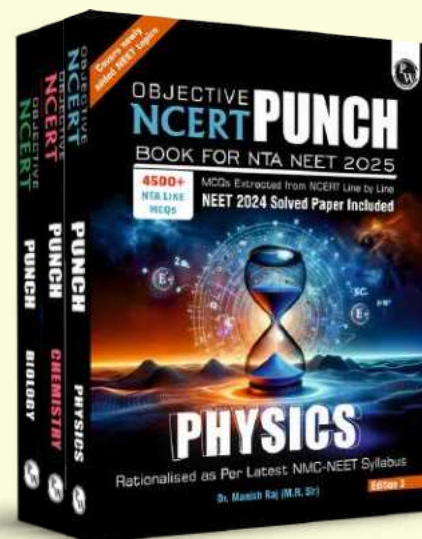
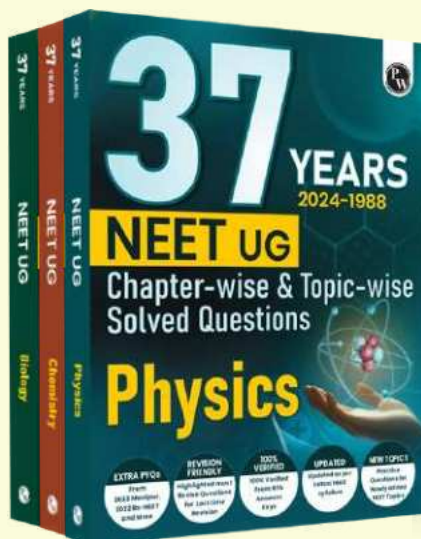
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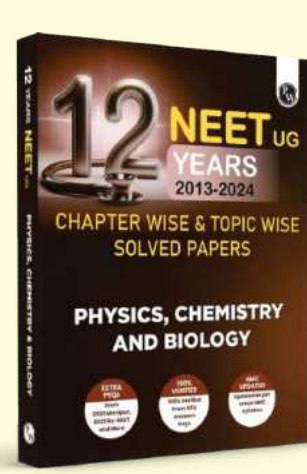
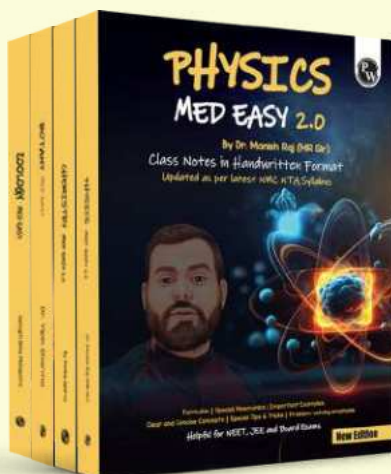
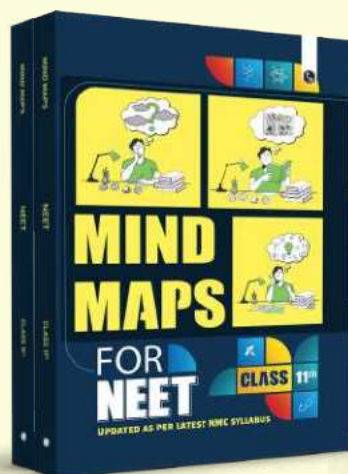
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